THE MECHANICS OF LINEAR SPATIAL FILTERING

A linear spatial filter performs a sum-of-products operation between an image f and a *filter kernel*, w. The kernel is an array whose size defines the neighborhood of operation, and whose coefficients determine the nature of the filter. Other terms used to refer to a spatial filter kernel are *mask*, *template*, and *window*. We use the term *filter kernel* or simply *kernel*.

Figure 3.28 illustrates the mechanics of linear spatial filtering using a 3×3 kernel. At any point (x, y) in the image, the response, $g(x, y)$, of the filter is the sum of products of the kernel coefficients and the image pixels encompassed by the kernel:

$$\begin{aligned} g(x,y) = w(-1,-1)f(x-1,y-1) + w(-1,0)f(x-1,y) + \ldots \\ + w(0,0)f(x,y) + \ldots + w(1,1)f(x+1,y+1) \end{aligned} \tag{3-30}$$

As coordinates x and y are varied, the center of the kernel moves from pixel to pixel, generating the filtered image, g, in the process.[†]

Observe that the center coefficient of the kernel, $w(0,0)$, aligns with the pixel at location (x, y). For a kernel of size $m \times n$, we assume that $m = 2a + 1$ and $n = 2b + 1$, where a and b are nonnegative integers. This means that our focus is on kernels of odd size in both coordinate directions. In general, linear spatial filtering of an image of size $M \times N$ with a kernel of size $m \times n$ is given by the expression

It certainly is possible to work with kernels of even size, or mixed even and odd sizes. However, working with odd sizes simplifies indexing and is also more intuitive because the kernels have centers falling on integer values, and they are spatially symmetric.

$$g(x,y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s,t) f(x+s, y+t) \tag{3-31}$$

where x and y are varied so that the center (origin) of the kernel visits every pixel in f once. For a fixed value of (x, y), Eq. (3-31) implements the *sum of products* of the form shown in Eq. (3-30), but for a kernel of arbitrary odd size. As you will learn in the following section, this equation is a central tool in linear filtering.

SPATIAL CORRELATION AND CONVOLUTION

Spatial correlation is illustrated graphically in Fig. 3.28, and it is described mathematically by Eq. (3-31). Correlation consists of moving the center of a kernel over an image, and computing the sum of products at each location. The mechanics of *spatial convolution* are the same, except that the correlation kernel is rotated by 180°. Thus, when the values of a kernel are symmetric about its center, correlation and convolution yield the same result. The reason for rotating the kernel will become clear in the following discussion. The best way to explain the differences between the two concepts is by example.

We begin with a 1-D illustration, in which case Eq. (3-31) becomes

$$g(x) = \sum_{s=-a}^{a} w(s) f(x+s) \tag{3-32}$$

[†] A filtered pixel value typically is assigned to a corresponding location in a new image created to hold the results of filtering. It is seldom the case that filtered pixels replace the values of the corresponding location in the original image, as this would change the content of the image while filtering is being performed.

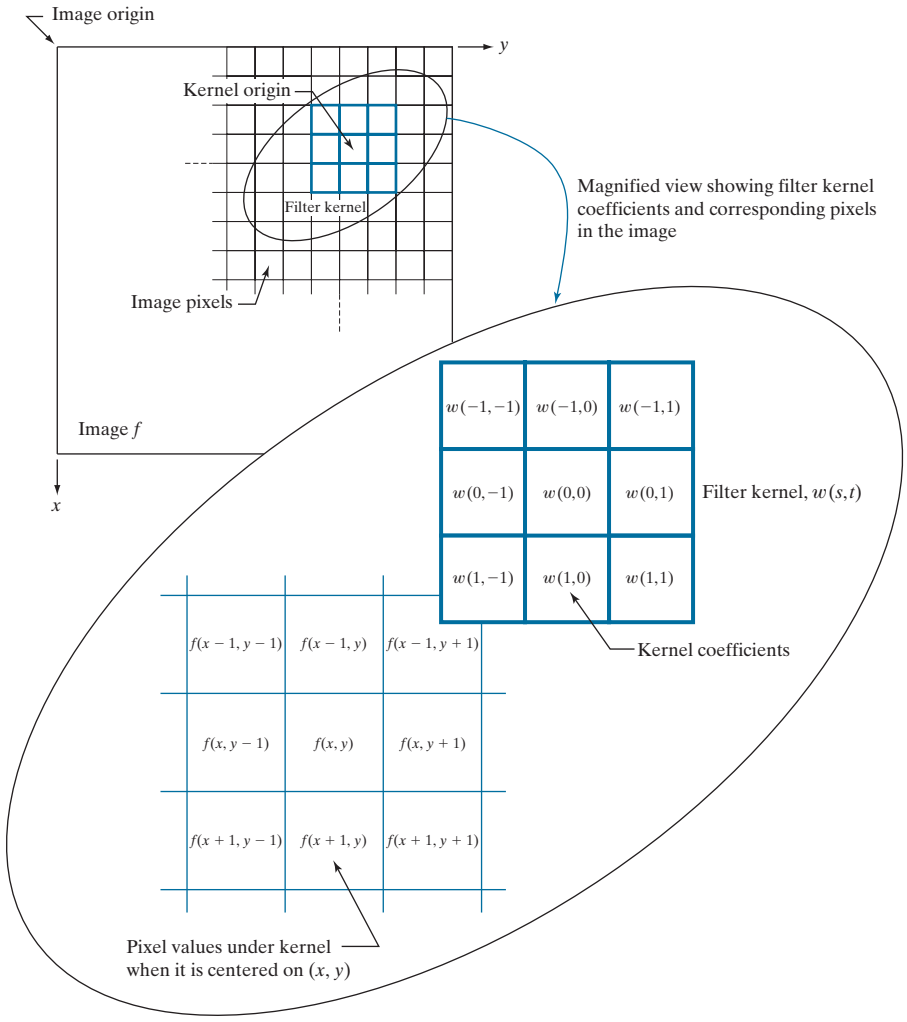


FIGURE 3.28 The mechanics of linear spatial filtering using a 3×3 kernel. The pixels are shown as squares to simplify the graphics. Note that the origin of the image is at the top left, but the origin of the kernel is at its center. Placing the origin at the center of spatially symmetric kernels simplifies writing expressions for linear filtering.

Figure 3.29(a) shows a 1-D function, f, and a kernel, w. The kernel is of size 1×5, so $a = 2$ and $b = 0$ in this case. Figure 3.29(b) shows the starting position used to perform correlation, in which w is positioned so that its center coefficient is coincident with the origin of f.

The first thing we notice is that part of w lies outside f, so the summation is undefined in that area. A solution to this problem is to *pad* function f with enough 0's on either side. In general, if the kernel is of size $1 \times m$, we need $(m-1)/2$ zeros on either side of f in order to handle the beginning and ending configurations of w with respect to f. Figure 3.29(c) shows a properly padded function. In this starting configuration, all coefficients of the kernel overlap valid values.

Zero padding is not the only padding option, as we will discuss in detail later in this chapter.

transition of the filter between low and high frequencies is instantaneous. Such filter functions are not realizable with physical components, and have issues with "ringing" when implemented digitally. However, ideal filters are very useful for illustrating numerous filtering phenomena, as you will learn in Chapter 4.

To lowpass-filter a spatial signal in the frequency domain, we first convert it to the frequency domain by computing its Fourier transform, and then *multiply* the result by the filter transfer function in Fig. 3.32(a) to eliminate frequency components with values higher than u_0. To return to the spatial domain, we take the inverse Fourier transform of the filtered signal. The result will be a blurred spatial domain function.

Because of the duality between the spatial and frequency domains, we can obtain the same result in the spatial domain by *convolving* the equivalent spatial domain filter kernel with the input spatial function. The equivalent spatial filter kernel is the inverse Fourier transform of the frequency-domain filter transfer function. Figure 3.32(b) shows the spatial filter kernel corresponding to the frequency domain filter transfer function in Fig. 3.32(a). The ringing characteristics of the kernel are evident in the figure. A central theme of digital filter design theory is obtaining faithful (and practical) approximations to the sharp cut off of ideal frequency domain filters while reducing their ringing characteristics.

A WORD ABOUT HOW SPATIAL FILTER KERNELS ARE CONSTRUCTED

We consider three basic approaches for constructing spatial filters in the following sections of this chapter. One approach is based on formulating filters based on mathematical properties. For example, a filter that computes the average of pixels in a neighborhood blurs an image. Computing an average is analogous to integration. Conversely, a filter that computes the local derivative of an image sharpens the image. We give numerous examples of this approach in the following sections.

A second approach is based on sampling a 2-D spatial function whose shape has a desired property. For example, we will show in the next section that samples from a Gaussian function can be used to construct a weighted-average (lowpass) filter. These 2-D spatial functions sometimes are generated as the inverse Fourier transform of 2-D filters specified in the frequency domain. We will give several examples of this approach in this and the next chapter.

A third approach is to design a spatial filter with a specified frequency response. This approach is based on the concepts discussed in the previous section, and falls in the area of digital filter design. A 1-D spatial filter with the desired response is obtained (typically using filter design software). The 1-D filter values can be expressed as a vector $\mathbf{v}$, and a 2-D separable kernel can then be obtained using Eq. (3-42). Or the 1-D filter can be rotated about its center to generate a 2-D kernel that approximates a circularly symmetric function. We will illustrate these techniques in Section 3.7.

3.5 SMOOTHING (LOWPASS) SPATIAL FILTERS

Smoothing (also called *averaging*) spatial filters are used to reduce sharp transitions in intensity. Because random noise typically consists of sharp transitions in

intensity, an obvious application of smoothing is noise reduction. Smoothing prior to image resampling to reduce aliasing, as will be discussed in Section 4.5, is also a common application. Smoothing is used to reduce irrelevant detail in an image, where "irrelevant" refers to pixel regions that are small with respect to the size of the filter kernel. Another application is for smoothing the false contours that result from using an insufficient number of intensity levels in an image, as discussed in Section 2.4. Smoothing filters are used in combination with other techniques for image enhancement, such as the histogram processing techniques discussed in Section 3.3, and unsharp masking, as discussed later in this chapter. We begin the discussion of smoothing filters by considering linear smoothing filters in some detail. We will introduce nonlinear smoothing filters later in this section.

As we discussed in Section 3.4, linear spatial filtering consists of convolving an image with a filter kernel. Convolving a smoothing kernel with an image blurs the image, with the degree of blurring being determined by the size of the kernel and the values of its coefficients. In addition to being useful in countless applications of image processing, lowpass filters are fundamental, in the sense that other important filters, including sharpening (highpass), bandpass, and bandreject filters, can be derived from lowpass filters, as we will show in Section 3.7.

We discuss in this section lowpass filters based on *box* and *Gaussian* kernels, both of which are separable. Most of the discussion will center on Gaussian kernels because of their numerous useful properties and breadth of applicability. We will introduce other smoothing filters in Chapters 4 and 5.

BOX FILTER KERNELS

The simplest, separable lowpass filter kernel is the *box kernel*, whose coefficients have the same value (typically 1). The name "box kernel" comes from a constant kernel resembling a box when viewed in 3-D. We showed a 3×3 box filter in Fig. 3.31(a). An $m \times n$ box filter is an $m \times n$ array of 1's, with a normalizing constant in front, whose value is 1 divided by the sum of the values of the coefficients (i.e., $1/mn$ when all the coefficients are 1's). This normalization, which we apply to all lowpass kernels, has two purposes. First, the average value of an area of constant intensity would equal that intensity in the filtered image, as it should. Second, normalizing the kernel in this way prevents introducing a *bias* during filtering; that is, the sum of the pixels in the original and filtered images will be the same (see Problem 3.31). Because in a box kernel all rows and columns are identical, the rank of these kernels is 1, which, as we discussed earlier, means that they are separable.

EXAMPLE 3.11: Lowpass filtering with a box kernel.

Figure 3.33(a) shows a test pattern image of size 1024×1024 pixels. Figures 3.33(b)-(d) are the results obtained using box filters of size $m \times m$ with $m = 3, 11$, and 21, respectively. For $m = 3$, we note a slight overall blurring of the image, with the image features whose sizes are comparable to the size of the kernel being affected significantly more. Such features include the thinner lines in the image and the noise pixels contained in the boxes on the right side of the image. The filtered image also has a thin gray border, the result of zero-padding the image prior to filtering. As indicated earlier, padding extends the boundaries of an image to avoid undefined operations when parts of a kernel lie outside the border of

or with strong geometrical components, the directionality of box filters often produces undesirable results. (Example 3.13 illustrates this issue.) These are but two applications in which box filters are not suitable.

The kernels of choice in applications such as those just mentioned are *circularly symmetric* (also called *isotropic*, meaning their response is independent of orientation). As it turns out, Gaussian kernels of the form

$$w(s,t) = G(s,t) = Ke^{-\frac{s^2+t^2}{2\sigma^2}} \tag{3-45}$$

Our interest here is strictly on the bell shape of the Gaussian function; thus, we dispense with the traditional multiplier of the Gaussian PDF and use a general constant, K, instead. Recall that σ controls the "spread" of a Gaussian function about its mean.

are the *only* circularly symmetric kernels that are also separable (Sahoo [1990]). Thus, because Gaussian kernels of this form are separable, Gaussian filters enjoy the same computational advantages as box filters, but have a host of additional properties that make them ideal for image processing, as you will learn in the following discussion. Variables s and t in Eq. (3-45), are real (typically discrete) numbers.

By letting $r = [s^2 + t^2]^{1/2}$ we can write Eq. (3-45) as

$$G(r) = Ke^{-\frac{r^2}{2\sigma^2}} \tag{3-46}$$

This equivalent form simplifies derivation of expressions later in this section. This form also reminds us that the function is circularly symmetric. Variable r is the distance from the center to any point on function G. Figure 3.34 shows values of r for several kernel sizes using integer values for s and t. Because we work generally with odd kernel sizes, the centers of such kernels fall on integer values, and it follows that all values of r^2 are integers also. You can see this by squaring the values in Fig. 3.34

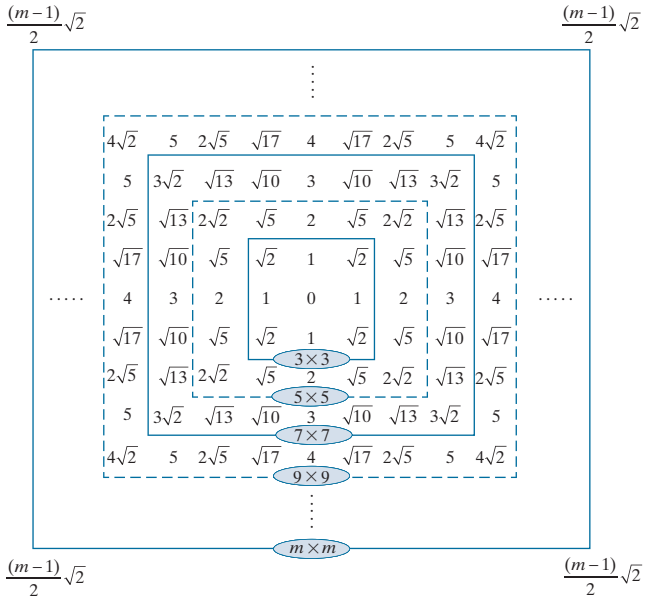


FIGURE 3.34 Distances from the center for various sizes of square kernels.

in computing the median). Median filters provide excellent noise reduction capabilities for certain types of random noise, with considerably less blurring than linear smoothing filters of similar size. Median filters are particularly effective in the presence of *impulse noise* (sometimes called *salt-and-pepper noise,* when it manisfests itself as white and black dots superimposed on an image).

The *median*, ξ, of a set of values is such that half the values in the set are less than or equal to ξ and half are greater than or equal to ξ. In order to perform median filtering at a point in an image, we first sort the values of the pixels in the neighborhood, determine their median, and assign that value to the pixel in the filtered image corresponding to the center of the neighborhood. For example, in a 3×3 neighborhood the median is the 5th largest value, in a 5×5 neighborhood it is the 13th largest value, and so on. When several values in a neighborhood are the same, all equal values are grouped. For example, suppose that a 3×3 neighborhood has values (10, 20, 20, 20, 15, 20, 20, 25, 100). These values are sorted as (10, 15, 20, 20, 20, 20, 20, 25, 100), which results in a median of 20. Thus, the principal function of median filters is to force points to be more like their neighbors. Isolated clusters of pixels that are light or dark with respect to their neighbors, and whose area is less than $m^2/2$ (one-half the filter area), are forced by an $m \times m$ median filter to have the value of the median intensity of the pixels in the neighborhood (see Problem 3.36).

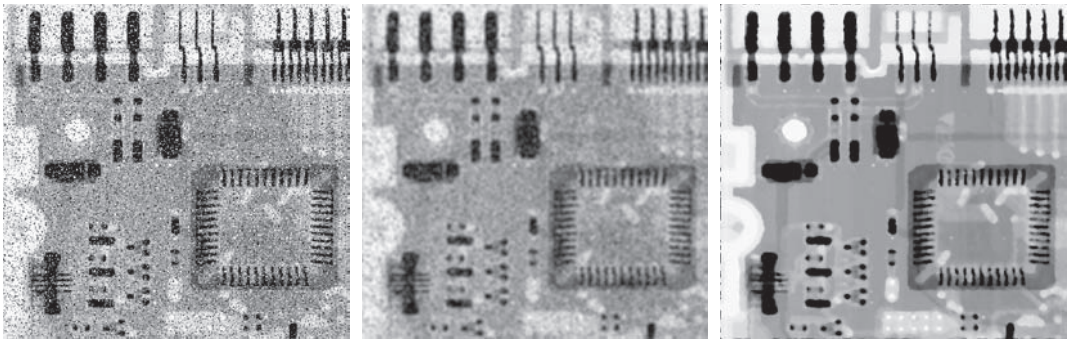
The median filter is by far the most useful order-statistic filter in image processing, but is not the only one. The median represents the 50th percentile of a ranked set of numbers, but ranking lends itself to many other possibilities. For example, using the 100th percentile results in the so-called *max filter*, which is useful for finding the brightest points in an image or for eroding dark areas adjacent to light regions. The response of a 3×3 max filter is given by $R = \max\{z_k \mid k = 1, 2, 3, \ldots, 9\}$. The 0th percentile filter is the *min filter*, used for the opposite purpose. Median, max, min, and several other nonlinear filters will be considered in more detail in Section 5.3.

EXAMPLE 3.17: Median filtering.

Figure 3.43(a) shows an X-ray image of a circuit board heavily corrupted by salt-and-pepper noise. To illustrate the superiority of median filtering over lowpass filtering in situations such as this, we show in Fig. 3.43(b) the result of filtering the noisy image with a Gaussian lowpass filter, and in Fig. 3.43(c) the result of using a median filter. The lowpass filter blurred the image and its noise reduction performance was poor. The superiority in all respects of median over lowpass filtering in this case is evident.

3.6 SHARPENING (HIGHPASS) SPATIAL FILTERS

Sharpening highlights transitions in intensity. Uses of image sharpening range from electronic printing and medical imaging to industrial inspection and autonomous guidance in military systems. In Section 3.5, we saw that image blurring could be accomplished in the spatial domain by pixel averaging (smoothing) in a neighborhood. Because averaging is analogous to integration, it is logical to conclude that sharpening can be accomplished by spatial differentiation. In fact, this is the case, and the following discussion deals with various ways of defining and implementing operators for sharpening by digital differentiation. The strength of the response of



a b c

FIGURE 3.43 (a) X-ray image of a circuit board, corrupted by salt-and-pepper noise. (b) Noise reduction using a 19×19 Gaussian lowpass filter kernel with $\sigma = 3$. (c) Noise reduction using a 7×7 median filter. (Original image courtesy of Mr. Joseph E. Pascente, Lixi, Inc.)

a derivative operator is proportional to the magnitude of the intensity discontinuity at the point at which the operator is applied. Thus, image differentiation enhances edges and other discontinuities (such as noise) and de-emphasizes areas with slowly varying intensities. As noted in Section 3.5, smoothing is often referred to as lowpass filtering, a term borrowed from frequency domain processing. In a similar manner, sharpening is often referred to as *highpass* filtering. In this case, high frequencies (which are responsible for fine details) are passed, while low frequencies are attenuated or rejected.

FOUNDATION

In the two sections that follow, we will consider in some detail sharpening filters that are based on first- and second-order derivatives, respectively. Before proceeding with that discussion, however, we stop to look at some of the fundamental properties of these derivatives in a digital context. To simplify the explanation, we focus attention initially on one-dimensional derivatives. In particular, we are interested in the behavior of these derivatives in areas of constant intensity, at the onset and end of discontinuities (*step* and *ramp discontinuities*), and along *intensity ramps*. As you will see in Chapter 10, these types of discontinuities can be used to model noise points, lines, and edges in an image.

Derivatives of a digital function are defined in terms of differences. There are various ways to define these differences. However, we require that any definition we use for a *first derivative:*

1. Must be zero in areas of constant intensity.
2. Must be nonzero at the onset of an intensity step or ramp.
3. Must be nonzero along intensity ramps.

Similarly, any definition of a *second derivative*

1. Must be zero in areas of constant intensity.
2. Must be nonzero at the onset *and* end of an intensity step or ramp.
3. Must be zero along intensity ramps.

We are dealing with digital quantities whose values are finite. Therefore, the maximum possible intensity change also is finite, and the shortest distance over which that change can occur is between adjacent pixels.

A basic definition of the *first-order* derivative of a one-dimensional function $f(x)$ is the difference

$$\frac{\partial f}{\partial x} = f(x+1) - f(x) \tag{3-48}$$

We will return to Eq. (3-48) in Section 10.2 and show how it follows from a Taylor series expansion. For now, we accept it as a definition.

We used a partial derivative here in order to keep the notation consistent when we consider an image function of two variables, $f(x, y)$, at which time we will be dealing with partial derivatives along the two spatial axes. Clearly, $\partial f/\partial x = df/dx$ when there is only one variable in the function; the same is true for the second derivative.

We define the *second-order* derivative of $f(x)$ as the difference

$$\frac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x) \tag{3-49}$$

These two definitions satisfy the conditions stated above, as we illustrate in Fig. 3.44, where we also examine the similarities and differences between first- and second-order derivatives of a digital function.

The values denoted by the small squares in Fig. 3.44(a) are the intensity values along a horizontal intensity profile (the dashed line connecting the squares is included to aid visualization). The actual numerical values of the scan line are shown inside the small boxes in 3.44(b). As Fig. 3.44(a) shows, the scan line contains three sections of constant intensity, an intensity ramp, and an intensity step. The circles indicate the onset or end of intensity transitions. The first- and second-order derivatives, computed using the two preceding definitions, are shown below the scan line values in Fig. 3.44(b), and are plotted in Fig. 3.44(c). When computing the first derivative at a location x, we subtract the value of the function at that location from the next point, as indicated in Eq. (3-48), so this is a "look-ahead" operation. Similarly, to compute the second derivative at x, we use the previous and the next points in the computation, as indicated in Eq. (3-49). To avoid a situation in which the previous or next points are outside the range of the scan line, we show derivative computations in Fig. 3.44 from the second through the penultimate points in the sequence.

As we traverse the profile from left to right we encounter first an area of constant intensity and, as Figs. 3.44(b) and (c) show, both derivatives are zero there, so condition (1) is satisfied by both. Next, we encounter an intensity ramp followed by a step, and we note that the first-order derivative is nonzero at the onset of the ramp and the step; similarly, the second derivative is nonzero at the onset and end of both the ramp and the step; therefore, property (2) is satisfied by both derivatives. Finally, we

a
b
c

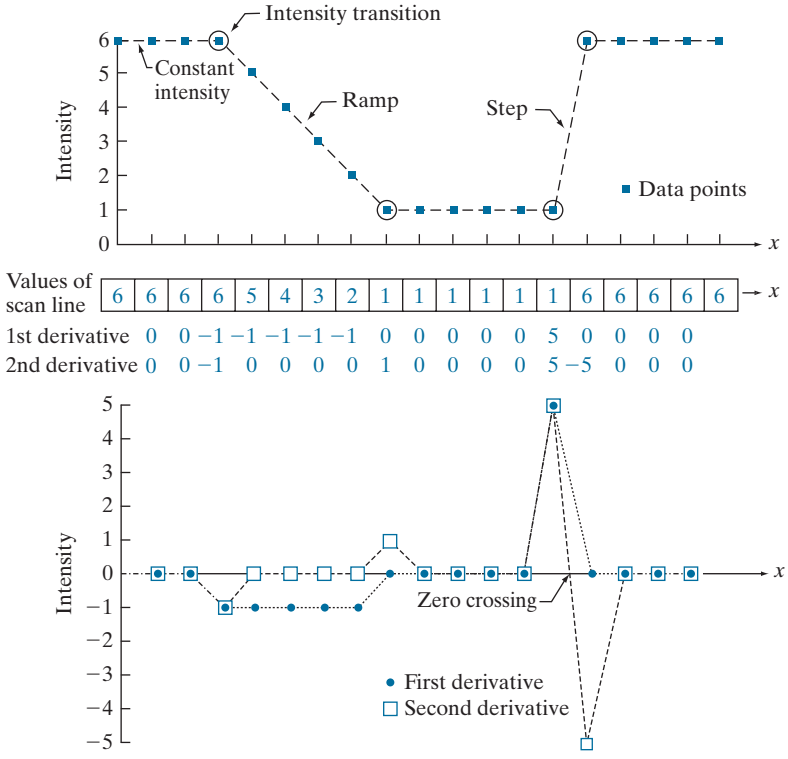


FIGURE 3.44 (a) A section of a horizontal scan line from an image, showing ramp and step edges, as well as constant segments. (b)Values of the scan line and its derivatives. (c) Plot of the derivatives, showing a zero crossing. In (a) and (c) points were joined by dashed lines as a visual aid.

see that property (3) is satisfied also by both derivatives because the first derivative is nonzero and the second is zero along the ramp. Note that the sign of the second derivative changes at the onset and end of a step or ramp. In fact, we see in Fig. 3.44(c) that in a step transition a line joining these two values crosses the horizontal axis midway between the two extremes. This *zero crossing* property is quite useful for locating edges, as you will see in Chapter 10.

Edges in digital images often are ramp-like transitions in intensity, in which case the first derivative of the image would result in thick edges because the derivative is nonzero along a ramp. On the other hand, the second derivative would produce a double edge one pixel thick, separated by zeros. From this, we conclude that the second derivative enhances fine detail much better than the first derivative, a property ideally suited for sharpening images. Also, second derivatives require fewer operations to implement than first derivatives, so our initial attention is on the former.

USING THE SECOND DERIVATIVE FOR IMAGE SHARPENING—THE LAPLACIAN

We will return to the second derivative in Chapter 10, where we use it extensively for image segmentation.

In this section we discuss the implementation of 2-D, second-order derivatives and their use for image sharpening. The approach consists of defining a discrete formulation of the second-order derivative and then constructing a filter kernel based on

that formulation. As in the case of Gaussian lowpass kernels in Section 3.5, we are interested here in isotropic kernels, whose response is independent of the direction of intensity discontinuities in the image to which the filter is applied.

It can be shown (Rosenfeld and Kak [1982]) that the simplest isotropic derivative operator (kernel) is the *Laplacian*, which, for a function (image) $f(x, y)$ of two variables, is defined as

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} \tag{3-50}$$

Because derivatives of any order are linear operations, the Laplacian is a linear operator. To express this equation in discrete form, we use the definition in Eq. (3-49), keeping in mind that we now have a second variable. In the x-direction, we have

$$\frac{\partial^2 f}{\partial x^2} = f(x+1, y) + f(x-1, y) - 2f(x, y) \tag{3-51}$$

and, similarly, in the y-direction, we have

$$\frac{\partial^2 f}{\partial y^2} = f(x, y+1) + f(x, y-1) - 2f(x, y) \tag{3-52}$$

It follows from the preceding three equations that the discrete Laplacian of two variables is

$$\nabla^2 f(x, y) = f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y) \tag{3-53}$$

This equation can be implemented using convolution with the kernel in Fig. 3.45(a); thus, the filtering mechanics for image sharpening are as described in Section 3.5 for lowpass filtering; we are simply using different coefficients here.

The kernel in Fig. 3.45(a) is isotropic for rotations in increments of 90° with respect to the x- and y-axes. The diagonal directions can be incorporated in the definition of the digital Laplacian by adding four more terms to Eq. (3-53). Because each diagonal term would contains a $-2f(x, y)$ term, the total subtracted from the difference terms

0	1	0
1	−4	1
0	1	0

1	1	1
1	−8	1
1	1	1

0	−1	0
−1	4	−1
0	−1	0

−1	−1	−1
−1	8	−1
−1	−1	−1

a b c d

FIGURE 3.45 (a) Laplacian kernel used to implement Eq. (3-53). (b) Kernel used to implement an extension of this equation that includes the diagonal terms. (c) and (d) Two other Laplacian kernels.

now would be $-8f(x, y)$. Figure 3.45(b) shows the kernel used to implement this new definition. This kernel yields isotropic results in increments of 45°. The kernels in Figs. 3.45(c) and (d) also are used to compute the Laplacian. They are obtained from definitions of the second derivatives that are the negatives of the ones we used here. They yield equivalent results, but the difference in sign must be kept in mind when combining a Laplacian-filtered image with another image.

Because the Laplacian is a derivative operator, it highlights sharp intensity transitions in an image and de-emphasizes regions of slowly varying intensities. This will tend to produce images that have grayish edge lines and other discontinuities, all superimposed on a dark, featureless background. Background features can be "recovered" while still preserving the sharpening effect of the Laplacian by adding the Laplacian image to the original. As noted in the previous paragraph, it is important to keep in mind which definition of the Laplacian is used. If the definition used has a negative center coefficient, then we *subtract* the Laplacian image from the original to obtain a sharpened result. Thus, the basic way in which we use the Laplacian for image sharpening is

$$g(x, y) = f(x, y) + c\left[\nabla^2 f(x, y)\right] \tag{3-54}$$

where $f(x, y)$ and $g(x, y)$ are the input and sharpened images, respectively. We let $c = -1$ if the Laplacian kernels in Fig. 3.45(a) or (b) is used, and $c = 1$ if either of the other two kernels is used.

EXAMPLE 3.18: Image sharpening using the Laplacian.

Figure 3.46(a) shows a slightly blurred image of the North Pole of the moon, and Fig. 3.46(b) is the result of filtering this image with the Laplacian kernel in Fig. 3.45(a) directly. Large sections of this image are black because the Laplacian image contains both positive and negative values, and all negative values are clipped at 0 by the display.

Figure 3.46(c) shows the result obtained using Eq. (3-54), with $c = -1$, because we used the kernel in Fig. 3.45(a) to compute the Laplacian. The detail in this image is unmistakably clearer and sharper than in the original image. Adding the Laplacian to the original image restored the overall intensity variations in the image. Adding the Laplacian increased the contrast at the locations of intensity discontinuities. The net result is an image in which small details were enhanced and the background tonality was reasonably preserved. Finally, Fig. 3.46(d) shows the result of repeating the same procedure but using the kernel in Fig. 3.45(b). Here, we note a significant improvement in sharpness over Fig. 3.46(c). This is not unexpected because using the kernel in Fig. 3.45(b) provides additional differentiation (sharpening) in the diagonal directions. Results such as those in Figs. 3.46(c) and (d) have made the Laplacian a tool of choice for sharpening digital images.

Because Laplacian images tend to be dark and featureless, a typical way to scale these images for display is to use Eqs. (2-31) and (2-32). This brings the most negative value to 0 and displays the full range of intensities. Figure 3.47 is the result of processing Fig. 3.46(b) in this manner. The dominant features of the image are edges and sharp intensity discontinuities. The background, previously black, is now gray as a result of scaling. This grayish appearance is typical of Laplacian images that have been scaled properly.

10.1 FUNDAMENTALS

Let R represent the entire spatial region occupied by an image. We may view image segmentation as a process that partitions R into n subregions, $R_1, R_2, \ldots, R_n$, such that

(a) $\bigcup_{i=1}^{n} R_i = R$.

(b) R_i is a connected set, for $i = 0, 1, 2, \ldots, n$.

(c) $R_i \cap R_j = \varnothing$ for all i and j, $i \neq j$.

(d) $Q(R_i)$ = TRUE for $i = 0, 1, 2, \ldots, n$.

(e) $Q(R_i \cup R_j)$ = FALSE for any adjacent regions R_i and R_j.

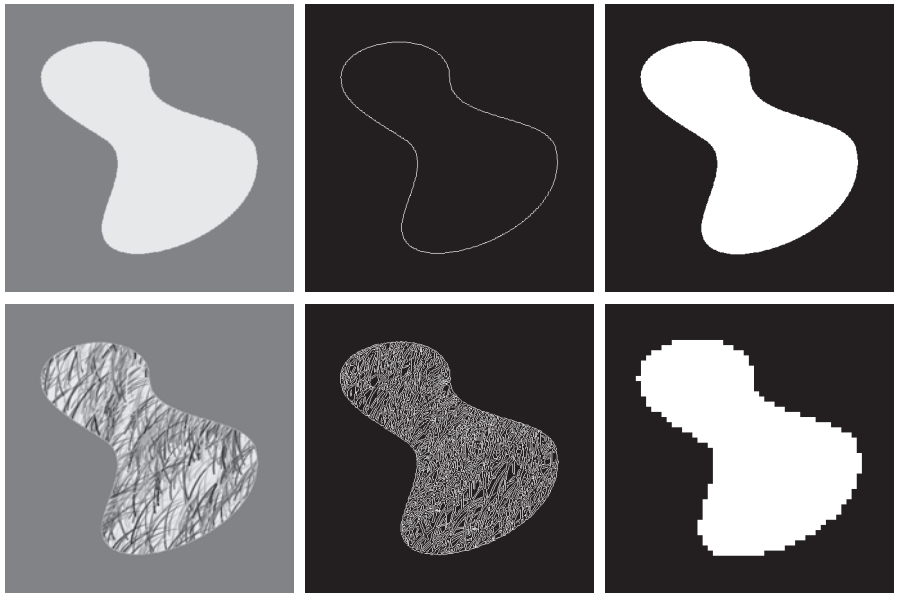
where $Q(R_k)$ is a logical predicate defined over the points in set R_k, and $\varnothing$ is the null set. The symbols $\cup$ and $\cap$ represent set union and intersection, respectively, as defined in Section 2.6. Two regions R_i and R_j are said to be *adjacent* if their union forms a connected set, as defined in Section 2.5. If the set formed by the union of two regions is not connected, the regions are said to *disjoint*.

Condition (a) indicates that the segmentation must be *complete*, in the sense that every pixel must be in a region. Condition (b) requires that points in a region be connected in some predefined sense (e.g., the points must be 8-connected). Condition (c) says that the regions must be disjoint. Condition (d) deals with the properties that must be satisfied by the pixels in a segmented region—for example, $Q(R_i)$ = TRUE if all pixels in R_i have the same intensity. Finally, condition (e) indicates that two adjacent regions R_i and R_j must be different in the sense of predicate Q.[†]

Thus, we see that the fundamental problem in segmentation is to partition an image into regions that satisfy the preceding conditions. Segmentation algorithms for monochrome images generally are based on one of two basic categories dealing with properties of intensity values: *discontinuity* and *similarity*. In the first category, we assume that boundaries of regions are sufficiently different from each other, and from the background, to allow boundary detection based on local discontinuities in intensity. *Edge-based* segmentation is the principal approach used in this category. *Region-based* segmentation approaches in the second category are based on partitioning an image into regions that are similar according to a set of predefined criteria.

Figure 10.1 illustrates the preceding concepts. Figure 10.1(a) shows an image of a region of constant intensity superimposed on a darker background, also of constant intensity. These two regions comprise the overall image. Figure 10.1(b) shows the result of computing the boundary of the inner region based on intensity discontinuities. Points on the inside and outside of the boundary are black (zero) because there are no discontinuities in intensity in those regions. To segment the image, we assign one level (say, white) to the pixels on or inside the boundary, and another level (e.g., black) to all points exterior to the boundary. Figure 10.1(c) shows the result of such a procedure. We see that conditions (a) through (c) stated at the beginning of this

[†] In general, Q can be a compound expression such as, "$Q(R_i)$ = TRUE if the average intensity of the pixels in region R_i is less than m_i AND if the standard deviation of their intensity is greater than σ_i," where m_i and σ_i are specified constants.



a b c
d e f

FIGURE 10.1
(a) Image of a constant intensity region. (b) Boundary based on intensity discontinuities. (c) Result of segmentation. (d) Image of a texture region. (e) Result of intensity discontinuity computations (note the large number of small edges). (f) Result of segmentation based on region properties.

section are satisfied by this result. The predicate of condition (d) is: If a pixel is on, or inside the boundary, label it white; otherwise, label it black. We see that this predicate is TRUE for the points labeled black or white in Fig. 10.1(c). Similarly, the two segmented regions (object and background) satisfy condition (e).

The next three images illustrate region-based segmentation. Figure 10.1(d) is similar to Fig. 10.1(a), but the intensities of the inner region form a textured pattern. Figure 10.1(e) shows the result of computing intensity discontinuities in this image. The numerous spurious changes in intensity make it difficult to identify a unique boundary for the original image because many of the nonzero intensity changes are connected to the boundary, so edge-based segmentation is not a suitable approach. However, we note that the outer region is constant, so all we need to solve this segmentation problem is a predicate that differentiates between textured and constant regions. The standard deviation of pixel values is a measure that accomplishes this because it is nonzero in areas of the texture region, and zero otherwise. Figure 10.1(f) shows the result of dividing the original image into subregions of size 8×8. Each subregion was then labeled white if the standard deviation of its pixels was positive (i.e., if the predicate was TRUE), and zero otherwise. The result has a "blocky" appearance around the edge of the region because groups of 8×8 squares were labeled with the same intensity (smaller squares would have given a smoother region boundary). Finally, note that these results also satisfy the five segmentation conditions stated at the beginning of this section.

10.2 POINT, LINE, AND EDGE DETECTION

The focus of this section is on segmentation methods that are based on detecting sharp, *local* changes in intensity. The three types of image characteristics in which

When we refer to lines, we are referring to thin structures, typically just a few pixels thick. Such lines may correspond, for example, to elements of a digitized architectural drawing, or roads in a satellite image.

we are interested are isolated points, lines, and edges. *Edge pixels* are pixels at which the intensity of an image changes abruptly, and *edges* (or *edge segments*) are sets of connected edge pixels (see Section 2.5 regarding connectivity). *Edge detectors* are local image processing tools designed to detect edge pixels. A *line* may be viewed as a (typically) thin edge segment in which the intensity of the background on either side of the line is either much higher or much lower than the intensity of the line pixels. In fact, as we will discuss later, lines give rise to so-called "roof edges." Finally, an *isolated point* may be viewed as a foreground (background) pixel surrounded by background (foreground) pixels.

BACKGROUND

As we saw in Section 3.5, local averaging smoothes an image. Given that averaging is analogous to integration, it is intuitive that abrupt, local changes in intensity can be detected using derivatives. For reasons that will become evident shortly, first- and second-order derivatives are particularly well suited for this purpose.

Derivatives of a digital function are defined in terms of *finite differences*. There are various ways to compute these differences but, as explained in Section 3.6, we require that any approximation used for first derivatives (1) must be zero in areas of constant intensity; (2) must be nonzero at the onset of an intensity step or ramp; and (3) must be nonzero at points along an intensity ramp. Similarly, we require that an approximation used for second derivatives (1) must be zero in areas of constant intensity; (2) must be nonzero at the onset and end of an intensity step or ramp; and (3) must be zero along intensity ramps. Because we are dealing with digital quantities whose values are finite, the maximum possible intensity change is also finite, and the shortest distance over which a change can occur is between adjacent pixels.

We obtain an approximation to the first-order derivative at an arbitrary point x of a one-dimensional function $f(x)$ by expanding the function $f(x+\Delta x)$ into a Taylor series about x

Remember, the notation $n!$ means "n factorial": $n! = 1\times2\times\cdots\times n$.

$$f(x+\Delta x) = f(x) + \Delta x\frac{\partial f(x)}{\partial x} + \frac{(\Delta x)^2}{2!}\frac{\partial^2 f(x)}{\partial x^2} + \frac{(\Delta x)^3}{3!}\frac{\partial^3 f(x)}{\partial x^3} + \cdots\cdots$$
$$= \sum_{n=0}^{\infty}\frac{(\Delta x)^n}{n!}\frac{\partial^n f(x)}{\partial x^n} \tag{10-1}$$

where Δx is the separation between samples of f. For our purposes, this separation is measured in pixel units. Thus, following the convention in the book, $\Delta x = 1$ for the sample preceding x and $\Delta x = -1$ for the sample following x. When $\Delta x = 1$, Eq. (10-1) becomes

Although this is an expression of only one variable, we used partial derivatives notation for consistency when we discuss functions of two variables later in this section.

$$f(x+1) = f(x) + \frac{\partial f(x)}{\partial x} + \frac{1}{2!}\frac{\partial^2 f(x)}{\partial x^2} + \frac{1}{3!}\frac{\partial^3 f(x)}{\partial x^3} + \cdots\cdots$$
$$= \sum_{n=0}^{\infty}\frac{1}{n!}\frac{\partial^n f(x)}{\partial x^n} \tag{10-2}$$

Similarly, when $\Delta x = -1$,

$$f(x-1) = f(x) - \frac{\partial f(x)}{\partial x} + \frac{1}{2!}\frac{\partial^2 f(x)}{\partial x^2} - \frac{1}{3!}\frac{\partial^3 f(x)}{\partial x^3} + \cdots\cdots$$
$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{n!}\frac{\partial^n f(x)}{\partial x^n} \tag{10-3}$$

In what follows, we compute *intensity differences* using just a few terms of the Taylor series. For first-order derivatives we use only the linear terms, and we can form differences in one of three ways.

The *forward difference* is obtained from Eq. (10-2):

$$\frac{\partial f(x)}{\partial x} = f'(x) = f(x+1) - f(x) \tag{10-4}$$

where, as you can see, we kept only the linear terms. The *backward difference* is similarly obtained by keeping only the linear terms in Eq. (10-3):

$$\frac{\partial f(x)}{\partial x} = f'(x) = f(x) - f(x-1) \tag{10-5}$$

and the *central difference* is obtained by subtracting Eq. (10-3) from Eq. (10-2):

$$\frac{\partial f(x)}{\partial x} = f'(x) = \frac{f(x+1) - f(x-1)}{2} \tag{10-6}$$

The higher terms of the series that we did not use represent the error between an exact and an approximate derivative expansion. In general, the more terms we use from the Taylor series to represent a derivative, the more accurate the approximation will be. To include more terms implies that more points are used in the approximation, yielding a lower error. However, it turns out that central differences have a lower error for the same number of points (see Problem 10.1). For this reason, derivatives are usually expressed as central differences.

The *second order* derivative based on a central difference, $\partial^2 f(x)/\partial x^2$, is obtained by adding Eqs. (10-2) and (10-3):

$$\frac{\partial^2 f(x)}{\partial x^2} = f''(x) = f(x+1) - 2f(x) + f(x-1) \tag{10-7}$$

To obtain the *third order, central derivative* we need one more point on either side of x. That is, we need the Taylor expansions for $f(x+2)$ and $f(x-2)$, which we obtain from Eqs. (10-2) and (10-3) with $\Delta x = 2$ and $\Delta x = -2$, respectively. The strategy is to combine the two Taylor expansions to eliminate all derivatives lower than the third. The result after ignoring all higher-order terms [see Problem 10.2(a)] is

$$\frac{\partial^3 f(x)}{\partial x^3} = f'''(x) = \frac{f(x+2) - 2f(x+1) + 0f(x) + 2f(x-1) - f(x-2)}{2} \quad (10\text{-}8)$$

Similarly [see Problem 10.2(b)], the *fourth* finite difference (the highest we use in the book) after ignoring all higher order terms is given by

$$\frac{\partial^4 f(x)}{\partial x^4} = f''''(x) = f(x+2) - 4f(x+1) + 6f(x) - 4f(x-1) + f(x-2) \quad (10\text{-}9)$$

Table 10.1 summarizes the first four central derivatives just discussed. Note the symmetry of the coefficients about the center point. This symmetry is at the root of why central differences have a lower approximation error for the same number of points than the other two differences. For two variables, we apply the results in Table 10.1 to each variable independently. For example,

$$\frac{\partial^2 f(x,y)}{\partial x^2} = f(x+1,y) - 2f(x,y) + f(x-1,y) \quad (10\text{-}10)$$

and

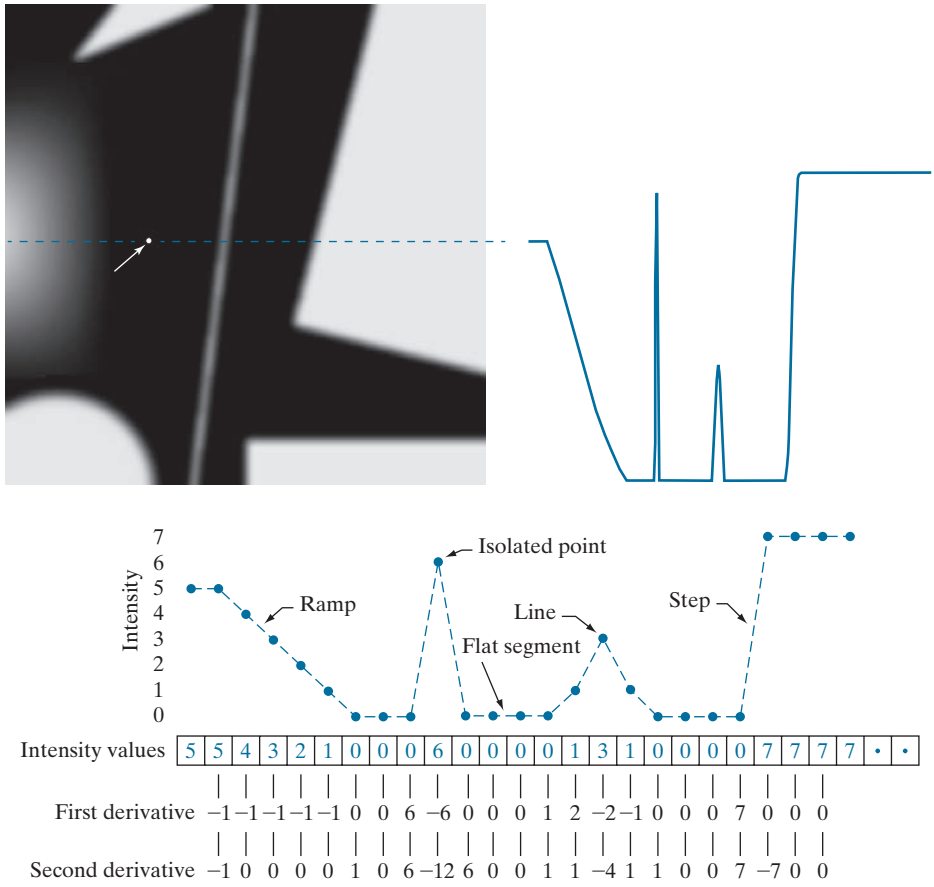
$$\frac{\partial^2 f(x,y)}{\partial y^2} = f(x,y+1) - 2f(x,y) + f(x,y-1) \quad (10\text{-}11)$$

It is easily verified that the first and second-order derivatives in Eqs. (10-4) through (10-7) satisfy the conditions stated at the beginning of this section regarding derivatives of the first and second order. To illustrate this, consider Fig. 10.2. Part (a) shows an image of various objects, a line, and an isolated point. Figure 10.2(b) shows a horizontal intensity profile (scan line) through the center of the image, including the isolated point. Transitions in intensity between the solid objects and the background along the scan line show two types of edges: *ramp edges* (on the left) and *step edges* (on the right). As we will discuss later, intensity transitions involving thin objects such as lines often are referred to as *roof edges*.

Figure 10.2(c) shows a simplified profile, with just enough points to make it possible for us to analyze manually how the first- and second-order derivatives behave as they encounter a point, a line, and the edges of objects. In this diagram the transition

TABLE 10.1
First four central digital derivatives (finite differences) for samples taken uniformly, $\Delta x = 1$ units apart.

	$f(x+2)$	$f(x+1)$	$f(x)$	$f(x-1)$	$f(x-2)$
$2f'(x)$		1	0	–1	
$f''(x)$		1	–2	1	
$2f'''(x)$	1	–2	0	2	–1
$f''''(x)$	1	–4	6	–4	1



a b
c

FIGURE 10.2
(a) Image.
(b) Horizontal intensity profile that includes the isolated point indicated by the arrow.
(c) Subsampled profile; the dashes were added for clarity. The numbers in the boxes are the intensity values of the dots shown in the profile. The derivatives were obtained using Eqs. (10-4) for the first derivative and Eq. (10-7) for the second.

in the ramp spans four pixels, the noise point is a single pixel, the line is three pixels thick, and the transition of the step edge takes place between adjacent pixels. The number of intensity levels was limited to eight for simplicity.

Consider the properties of the first and second derivatives as we traverse the profile from left to right. Initially, the first-order derivative is nonzero at the onset and along the entire intensity ramp, while the second-order derivative is nonzero only at the onset and end of the ramp. Because the edges of digital images resemble this type of transition, we conclude that first-order derivatives produce "thick" edges, and second-order derivatives much thinner ones. Next we encounter the isolated noise point. Here, the magnitude of the response at the point is much stronger for the second- than for the first-order derivative. This is not unexpected, because a second-order derivative is much more aggressive than a first-order derivative in enhancing sharp changes. Thus, we can expect second-order derivatives to enhance fine detail (including noise) much more than first-order derivatives. The line in this example is rather thin, so it too is fine detail, and we see again that the second derivative has a larger magnitude. Finally, note in both the ramp and step edges that the

w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

FIGURE 10.3 A general 3×3 spatial filter kernel. The w's are the kernel coefficients (weights).

second derivative has opposite signs (negative to positive or positive to negative) as it transitions into and out of an edge. This "double-edge" effect is an important characteristic that can be used to locate edges, as we will show later in this section. As we move into the edge, the sign of the second derivative is used also to determine whether an edge is a transition from light to dark (negative second derivative), or from dark to light (positive second derivative)

In summary, we arrive at the following conclusions: (1) First-order derivatives generally produce thicker edges. (2) Second-order derivatives have a stronger response to fine detail, such as thin lines, isolated points, and noise. (3) Second-order derivatives produce a double-edge response at ramp and step transitions in intensity. (4) The sign of the second derivative can be used to determine whether a transition into an edge is from light to dark or dark to light.

The approach of choice for computing first and second derivatives at every pixel location in an image is to use spatial convolution. For the 3×3 filter kernel in Fig. 10.3, the procedure is to compute the sum of products of the kernel coefficients with the intensity values in the region encompassed by the kernel, as we explained in Section 3.4. That is, the response of the filter at the center point of the kernel is

This equation is an expansion of Eq. (3-35) for a 3×3 kernel, valid at one point, and using simplified subscript notation for the kernel coefficients.

$$\begin{aligned} Z &= w_1 z_1 + w_2 z_2 + \ldots + w_9 z_9 \\ &= \sum_{k=1}^{9} w_k z_k \end{aligned} \tag{10-12}$$

where z_k is the intensity of the pixel whose spatial location corresponds to the location of the kth kernel coefficient.

DETECTION OF ISOLATED POINTS

Based on the conclusions reached in the preceding section, we know that point detection should be based on the second derivative which, from the discussion in Section 3.6, means using the Laplacian:

$$\nabla^2 f(x, y) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} \tag{10-13}$$

where the partial derivatives are computed using the second-order finite differences in Eqs. (10-10) and (10-11). The Laplacian is then

$$\nabla^2 f(x,y) = f(x+1,y) + f(x-1,y) + f(x,y+1) + f(x,y-1) - 4f(x,y) \quad (10\text{-}14)$$

As explained in Section 3.6, this expression can be implemented using the Laplacian kernel in Fig. 10.4(a) in Example 10.1. We then we say that a point has been detected at a location (x,y) on which the kernel is centered if the absolute value of the response of the filter at that point exceeds a specified threshold. Such points are labeled 1 and all others are labeled 0 in the output image, thus producing a binary image. In other words, we use the expression:

$$g(x,y) = \begin{cases} 1 & \text{if } |Z(x,y)| > T \\ 0 & \text{otherwise} \end{cases} \quad (10\text{-}15)$$

where $g(x,y)$ is the output image, T is a nonnegative threshold, and Z is given by Eq. (10-12). This formulation simply measures the weighted differences between a pixel and its 8-neighbors. Intuitively, the idea is that the intensity of an isolated point will be quite different from its surroundings, and thus will be easily detectable by this type of kernel. Differences in intensity that are considered of interest are those large enough (as determined by T) to be considered isolated points. Note that, as usual for a derivative kernel, the coefficients sum to zero, indicating that the filter response will be zero in areas of constant intensity.

EXAMPLE 10.1: Detection of isolated points in an image.

Figure 10.4(b) is an X-ray image of a turbine blade from a jet engine. The blade has a porosity manifested by a single black pixel in the upper-right quadrant of the image. Figure 10.4(c) is the result of filtering the image with the Laplacian kernel, and Fig. 10.4(d) shows the result of Eq. (10-15) with T equal to 90% of the highest absolute pixel value of the image in Fig. 10.4(c). The single pixel is clearly visible in this image at the tip of the arrow (the pixel was enlarged to enhance its visibility). This type of detection process is specialized because it is based on abrupt intensity changes at single-pixel locations that are surrounded by a homogeneous background in the area of the detector kernel. When this condition is not satisfied, other methods discussed in this chapter are more suitable for detecting intensity changes.

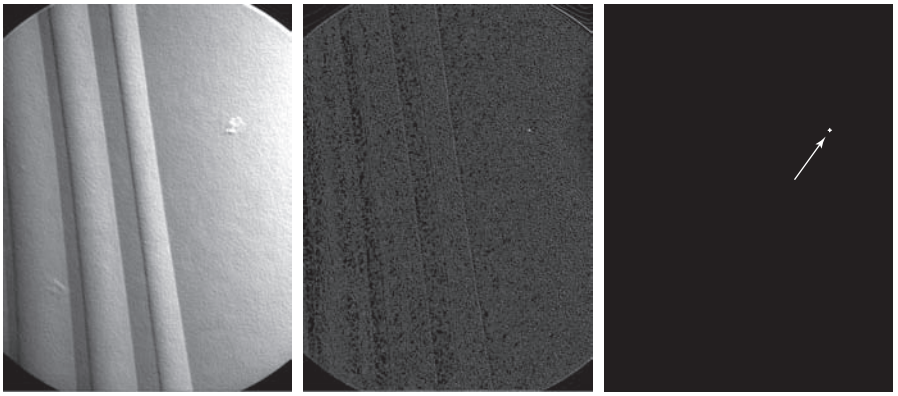
LINE DETECTION

The next level of complexity is line detection. Based on the discussion earlier in this section, we know that for line detection we can expect second derivatives to result in a stronger filter response, and to produce thinner lines than first derivatives. Thus, we can use the Laplacian kernel in Fig. 10.4(a) for line detection also, keeping in mind that the double-line effect of the second derivative must be handled properly. The following example illustrates the procedure.

a
b c d

FIGURE 10.4
(a) Laplacian kernel used for point detection. (b) X-ray image of a turbine blade with a porosity manifested by a single black pixel. (c) Result of convolving the kernel with the image. (d) Result of using Eq. (10-15) was a single point (shown enlarged at the tip of the arrow). (Original image courtesy of X-TEK Systems, Ltd.)

1	1	1
1	−8	1
1	1	1

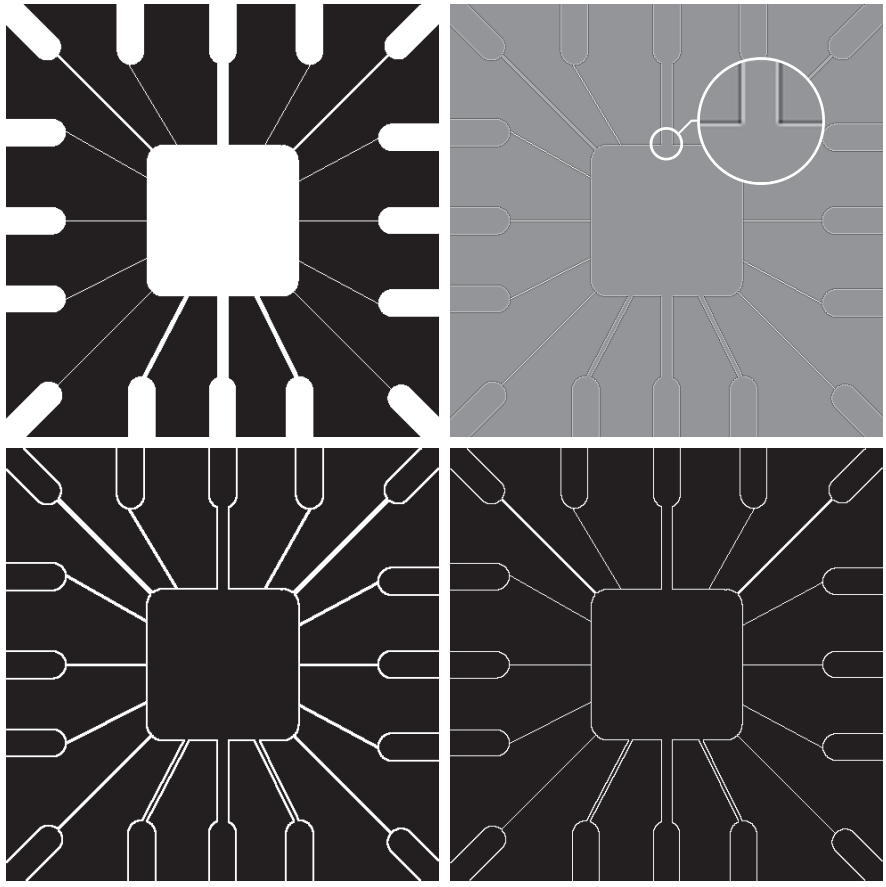


EXAMPLE 10.2: Using the Laplacian for line detection.

Figure 10.5(a) shows a 486×486 (binary) portion of a wire-bond mask for an electronic circuit, and Fig. 10.5(b) shows its Laplacian image. Because the Laplacian image contains negative values (see the discussion after Example 3.18), scaling is necessary for display. As the magnified section shows, mid gray represents zero, darker shades of gray represent negative values, and lighter shades are positive. The double-line effect is clearly visible in the magnified region.

At first, it might appear that the negative values can be handled simply by taking the absolute value of the Laplacian image. However, as Fig. 10.5(c) shows, this approach doubles the thickness of the lines. A more suitable approach is to use only the positive values of the Laplacian (in noisy situations we use the values that exceed a positive threshold to eliminate random variations about zero caused by the noise). As Fig. 10.5(d) shows, this approach results in thinner lines that generally are more useful. Note in Figs. 10.5(b) through (d) that when the lines are wide with respect to the size of the Laplacian kernel, the lines are separated by a zero "valley." This is not unexpected. For example, when the 3×3 kernel is centered on a line of constant intensity 5 pixels wide, the response will be zero, thus producing the effect just mentioned. When we talk about line detection, the assumption is that lines are thin with respect to the size of the detector. Lines that do not satisfy this assumption are best treated as regions and handled by the edge detection methods discussed in the following section.

The Laplacian detector kernel in Fig. 10.4(a) is isotropic, so its response is independent of direction (with respect to the four directions of the 3×3 kernel: vertical, horizontal, and two diagonals). Often, interest lies in detecting lines in *specified*

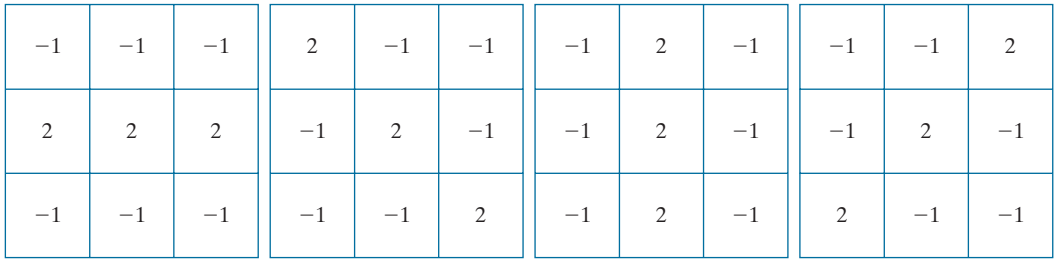


a b
c d

FIGURE 10.5
(a) Original image.
(b) Laplacian image; the magnified section shows the positive/negative double-line effect characteristic of the Laplacian.
(c) Absolute value of the Laplacian.
(d) Positive values of the Laplacian.

directions. Consider the kernels in Fig. 10.6. Suppose that an image with a constant background and containing various lines (oriented at 0°, ±45°, and 90°) is filtered with the first kernel. The maximum responses would occur at image locations in which a horizontal line passes through the middle row of the kernel. This is easily verified by sketching a simple array of 1's with a line of a different intensity (say, 5s) running horizontally through the array. A similar experiment would reveal that the second kernel in Fig. 10.6 responds best to lines oriented at +45°; the third kernel to vertical lines; and the fourth kernel to lines in the −45° direction. The preferred direction of each kernel is weighted with a larger coefficient (i.e., 2) than other possible directions. The coefficients in each kernel sum to zero, indicating a zero response in areas of constant intensity.

Let Z_1, Z_2, Z_3, and Z_4 denote the responses of the kernels in Fig. 10.6, from left to right, where the Zs are given by Eq. (10-12). Suppose that an image is filtered with these four kernels, one at a time. If, at a given point in the image, $|Z_k| > |Z_j|$, for all $j \neq k$, that point is said to be more likely associated with a line in the direction of kernel k. For example, if at a point in the image, $|Z_1| > |Z_j|$ for $j = 2,3,4$, that



a b c d

FIGURE 10.6 Line detection kernels. Detection angles are with respect to the axis system in Fig. 2.19, with positive angles measured counterclockwise with respect to the (vertical) x-axis.

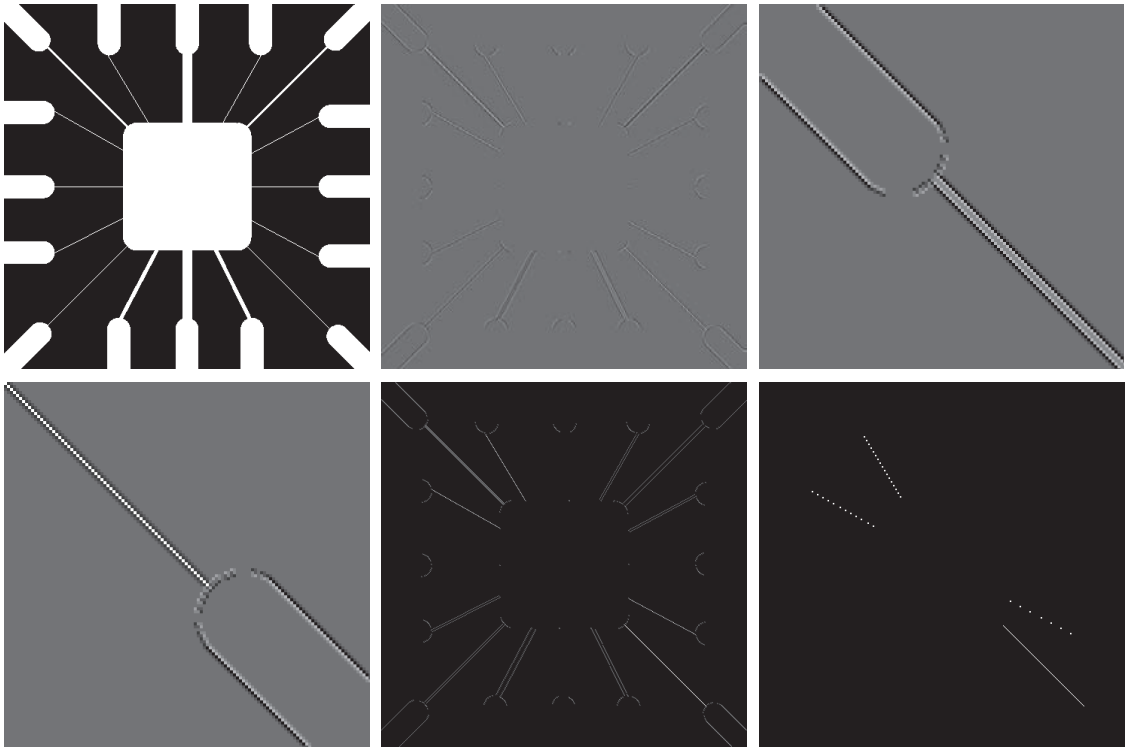
point is said to be more likely associated with a horizontal line. If we are interested in detecting all the lines in an image in the direction defined by a given kernel, we simply run the kernel through the image and threshold the absolute value of the result, as in Eq. (10-15). The nonzero points remaining after thresholding are the strongest responses which, for lines one pixel thick, correspond closest to the direction defined by the kernel. The following example illustrates this procedure.

EXAMPLE 10.3: Detecting lines in specified directions.

Figure 10.7(a) shows the image used in the previous example. Suppose that we are interested in finding all the lines that are one pixel thick and oriented at +45°. For this purpose, we use the kernel in Fig. 10.6(b). Figure 10.7(b) is the result of filtering the image with that kernel. As before, the shades darker than the gray background in Fig. 10.7(b) correspond to negative values. There are two principal segments in the image oriented in the +45° direction, one in the top left and one at the bottom right. Figures 10.7(c) and (d) show zoomed sections of Fig. 10.7(b) corresponding to these two areas. The straight line segment in Fig. 10.7(d) is brighter than the segment in Fig. 10.7(c) because the line segment in the bottom right of Fig. 10.7(a) is one pixel thick, while the one at the top left is not. The kernel is "tuned" to detect one-pixel-thick lines in the +45° direction, so we expect its response to be stronger when such lines are detected. Figure 10.7(e) shows the positive values of Fig. 10.7(b). Because we are interested in the strongest response, we let T equal 254 (the maximum value in Fig. 10.7(e) minus one). Figure 10.7(f) shows in white the points whose values satisfied the condition $g > T$, where g is the image in Fig. 10.7(e). The isolated points in the figure are points that also had similarly strong responses to the kernel. In the original image, these points and their immediate neighbors are oriented in such a way that the kernel produced a maximum response at those locations. These isolated points can be detected using the kernel in Fig. 10.4(a) and then deleted, or they can be deleted using morphological operators, as discussed in the last chapter.

EDGE MODELS

Edge detection is an approach used frequently for segmenting images based on abrupt (local) changes in intensity. We begin by introducing several ways to model edges and then discuss a number of approaches for edge detection.



a b c
d e f

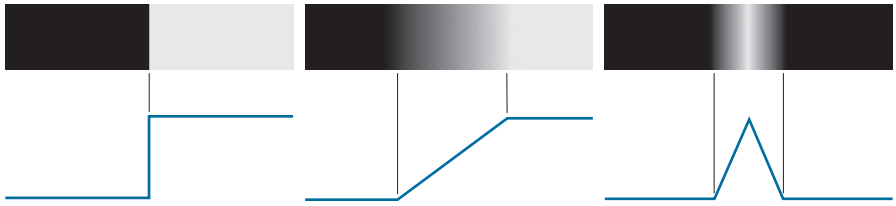
FIGURE 10.7 (a) Image of a wire-bond template. (b) Result of processing with the +45° line detector kernel in Fig. 10.6. (c) Zoomed view of the top left region of (b). (d) Zoomed view of the bottom right region of (b). (e) The image in (b) with all negative values set to zero. (f) All points (in white) whose values satisfied the condition $g > T$, where g is the image in (e) and $T = 254$ (the maximum pixel value in the image minus 1). (The points in (f) were enlarged to make them easier to see.)

Edge models are classified according to their intensity profiles. A *step edge* is characterized by a transition between two intensity levels occurring ideally over the distance of one pixel. Figure 10.8(a) shows a section of a vertical step edge and a horizontal intensity profile through the edge. Step edges occur, for example, in images generated by a computer for use in areas such as solid modeling and animation. These clean, *ideal* edges can occur over the distance of one pixel, provided that no additional processing (such as smoothing) is used to make them look "real." Digital step edges are used frequently as edge models in algorithm development. For example, the Canny edge detection algorithm discussed later in this section was derived originally using a step-edge model.

In practice, digital images have edges that are blurred and noisy, with the degree of blurring determined principally by limitations in the focusing mechanism (e.g., lenses in the case of optical images), and the noise level determined principally by the electronic components of the imaging system. In such situations, edges are more

a b c

FIGURE 10.8 From left to right, models (ideal representations) of a step, a ramp, and a roof edge, and their corresponding intensity profiles.



closely modeled as having an intensity *ramp* profile, such as the edge in Fig. 10.8(b). The slope of the ramp is inversely proportional to the degree to which the edge is blurred. In this model, we no longer have a single "edge point" along the profile. Instead, an edge point now is any point contained in the ramp, and an edge segment would then be a set of such points that are connected.

A third type of edge is the so-called *roof edge*, having the characteristics illustrated in Fig. 10.8(c). Roof edges are models of lines through a region, with the base (width) of the edge being determined by the thickness and sharpness of the line. In the limit, when its base is one pixel wide, a roof edge is nothing more than a one-pixel-thick line running through a region in an image. Roof edges arise, for example, in range imaging, when thin objects (such as pipes) are closer to the sensor than the background (such as walls). The pipes appear brighter and thus create an image similar to the model in Fig. 10.8(c). Other areas in which roof edges appear routinely are in the digitization of line drawings and also in satellite images, where thin features, such as roads, can be modeled by this type of edge.

It is not unusual to find images that contain all three types of edges. Although blurring and noise result in deviations from the ideal shapes, edges in images that are reasonably sharp and have a moderate amount of noise do resemble the characteristics of the edge models in Fig. 10.8, as the profiles in Fig. 10.9 illustrate. What the models in Fig. 10.8 allow us to do is write mathematical expressions for edges in the development of image processing algorithms. The performance of these algorithms will depend on the differences between actual edges and the models used in developing the algorithms.

Figure 10.10(a) shows the image from which the segment in Fig. 10.8(b) was extracted. Figure 10.10(b) shows a horizontal intensity profile. This figure shows also the first and second derivatives of the intensity profile. Moving from left to right along the intensity profile, we note that the first derivative is positive at the onset of the ramp and at points on the ramp, and it is zero in areas of constant intensity. The second derivative is positive at the beginning of the ramp, negative at the end of the ramp, zero at points on the ramp, and zero at points of constant intensity. The signs of the derivatives just discussed would be reversed for an edge that transitions from light to dark. The intersection between the zero intensity axis and a line extending between the extrema of the second derivative marks a point called the *zero crossing* of the second derivative.

We conclude from these observations that the *magnitude* of the first derivative can be used to detect the presence of an edge at a point in an image. Similarly, the *sign* of the second derivative can be used to determine whether an edge pixel lies on

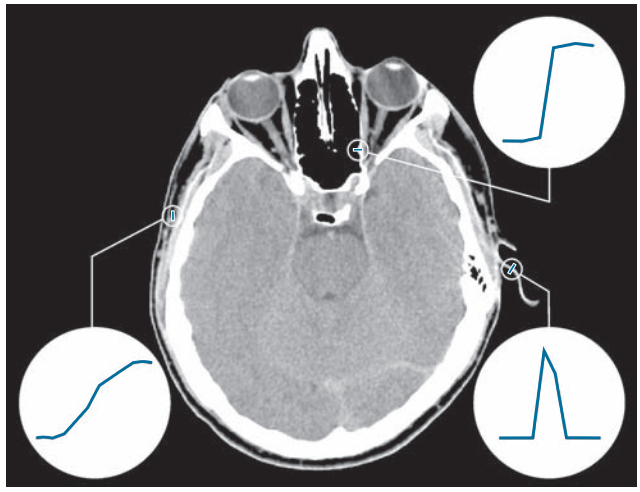
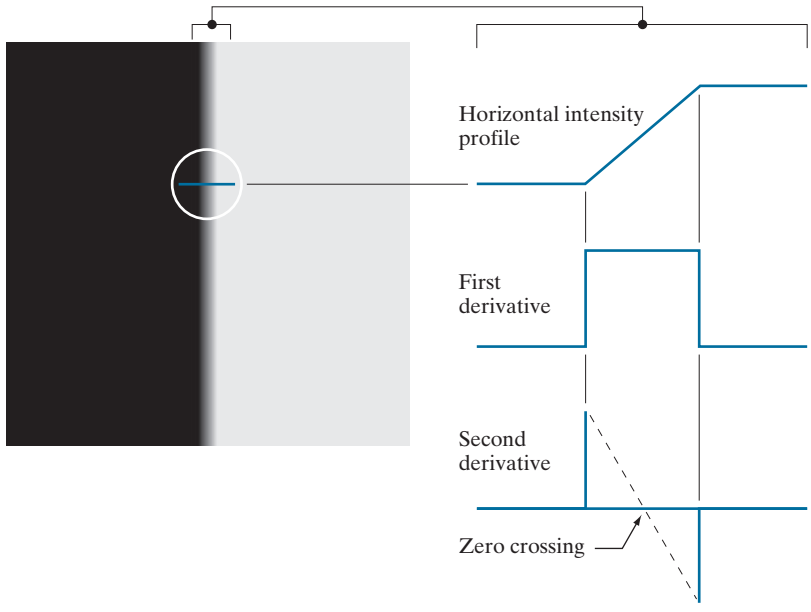


FIGURE 10.9 A 1508×1970 image showing (zoomed) actual ramp (bottom, left), step (top, right), and roof edge profiles. The profiles are from dark to light, in the areas enclosed by the small circles. The ramp and step profiles span 9 pixels and 2 pixels, respectively. The base of the roof edge is 3 pixels. (Original image courtesy of Dr. David R. Pickens, Vanderbilt University.)

the dark or light side of an edge. Two additional properties of the second derivative around an edge are: (1) it produces two values for every edge in an image; and (2) its zero crossings can be used for locating the centers of thick edges, as we will show later in this section. Some edge models utilize a smooth transition into and out of

a b

FIGURE 10.10 (a) Two regions of constant intensity separated by an ideal ramp edge. (b) Detail near the edge, showing a horizontal intensity profile, and its first and second derivatives.



the ramp (see Problem 10.9). However, the conclusions reached using those models are the same as with an ideal ramp, and working with the latter simplifies theoretical formulations. Finally, although attention thus far has been limited to a 1-D horizontal profile, a similar argument applies to an edge of any orientation in an image. We simply define a profile perpendicular to the edge direction at any desired point, and interpret the results in the same manner as for the vertical edge just discussed.

EXAMPLE 10.4: Behavior of the first and second derivatives in the region of a noisy edge.

The edge models in Fig. 10.8 are free of noise. The image segments in the first column in Fig. 10.11 show close-ups of four ramp edges that transition from a black region on the left to a white region on the right (keep in mind that the entire transition from black to white is a single edge). The image segment at the top left is free of noise. The other three images in the first column are corrupted by additive Gaussian noise with zero mean and standard deviation of 0.1, 1.0, and 10.0 intensity levels, respectively. The graph below each image is a horizontal intensity profile passing through the center of the image. All images have 8 bits of intensity resolution, with 0 and 255 representing black and white, respectively.

Consider the image at the top of the center column. As discussed in connection with Fig. 10.10(b), the derivative of the scan line on the left is zero in the constant areas. These are the two black bands shown in the derivative image. The derivatives at points on the ramp are constant and equal to the slope of the ramp. These constant values in the derivative image are shown in gray. As we move down the center column, the derivatives become increasingly different from the noiseless case. In fact, it would be difficult to associate the last profile in the center column with the first derivative of a ramp edge. What makes these results interesting is that the noise is almost visually undetectable in the images on the left column. These examples are good illustrations of the sensitivity of derivatives to noise.

As expected, the second derivative is even more sensitive to noise. The second derivative of the noiseless image is shown at the top of the right column. The thin white and black vertical lines are the positive and negative components of the second derivative, as explained in Fig. 10.10. The gray in these images represents zero (as discussed earlier, scaling causes zero to show as gray). The only noisy second derivative image that barely resembles the noiseless case corresponds to noise with a standard deviation of 0.1. The remaining second-derivative images and profiles clearly illustrate that it would be difficult indeed to detect their positive and negative components, which are the truly useful features of the second derivative in terms of edge detection.

The fact that such little visual noise can have such a significant impact on the two key derivatives used for detecting edges is an important issue to keep in mind. In particular, image smoothing should be a serious consideration prior to the use of derivatives in applications where noise with levels similar to those we have just discussed is likely to be present.

In summary, the three steps performed typically for edge detection are:

1. *Image smoothing for noise reduction.* The need for this step is illustrated by the results in the second and third columns of Fig. 10.11.
2. *Detection of edge points.* As mentioned earlier, this is a local operation that extracts from an image all points that are potential edge-point candidates.
3. *Edge localization.* The objective of this step is to select from the candidate points only the points that are members of the set of points comprising an edge.

The remainder of this section deals with techniques for achieving these objectives.

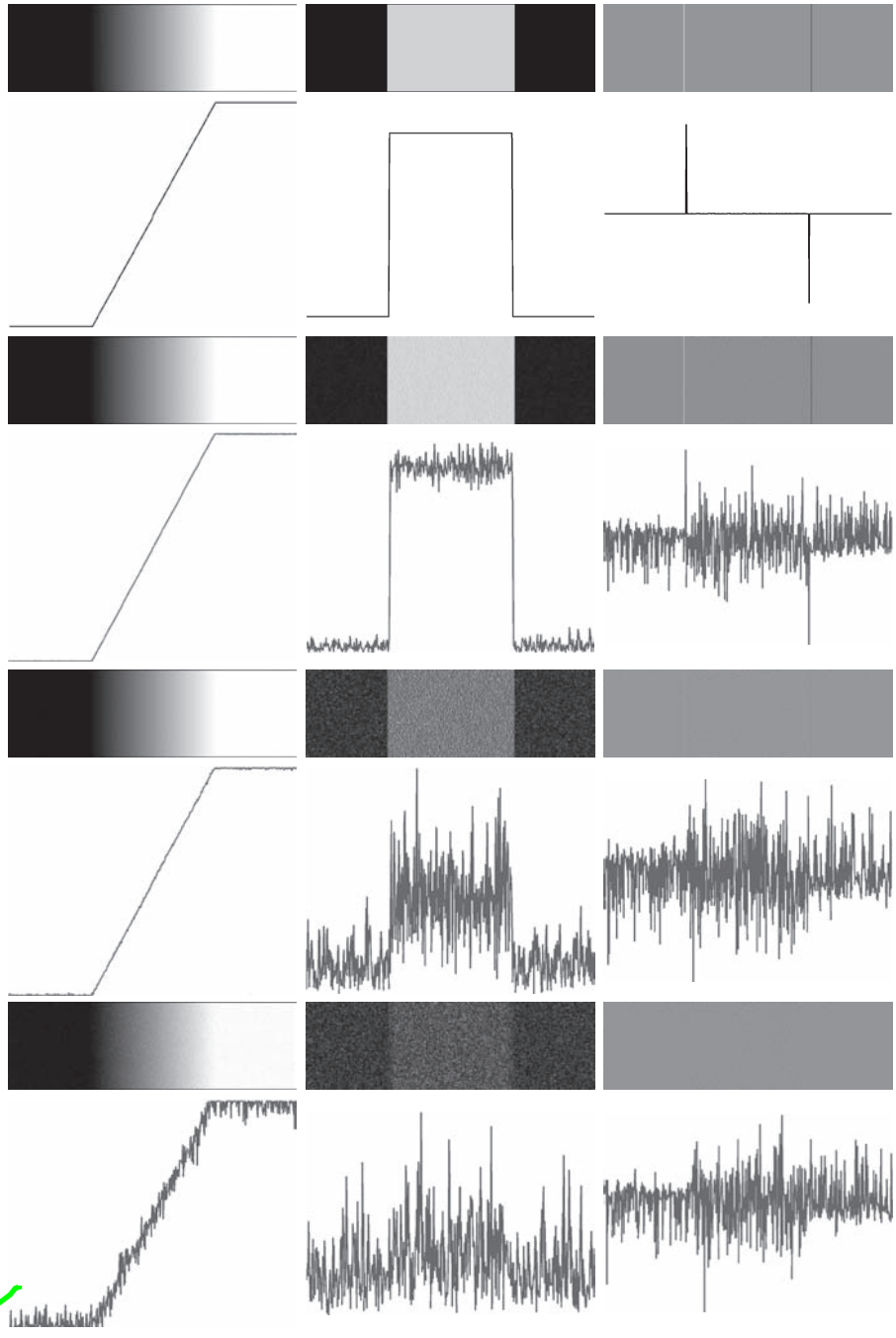


FIGURE 10.11 First column: 8-bit images with values in the range [0,255], and intensity profiles of a ramp edge corrupted by Gaussian noise of zero mean and standard deviations of 0.0, 0.1, 1.0, and 10.0 intensity levels, respectively. Second column: First-derivative images and intensity profiles. Third column: Second-derivative images and intensity profiles.

BASIC EDGE DETECTION

As illustrated in the preceding discussion, detecting changes in intensity for the purpose of finding edges can be accomplished using first- or second-order derivatives. We begin with first-order derivatives, and work with second-order derivatives in the following subsection.

The Image Gradient and Its Properties

The tool of choice for finding edge strength *and* direction at an arbitrary location (x, y) of an image, f, is the *gradient*, denoted by ∇f and defined as the *vector*

For convenience, we repeat here some of the gradient concepts and equations introduced in Chapter 3.

$$\nabla f(x,y) \equiv \text{grad}\left[f(x,y)\right] \equiv \begin{bmatrix} g_x(x,y) \\ g_y(x,y) \end{bmatrix} = \begin{bmatrix} \dfrac{\partial f(x,y)}{\partial x} \\ \dfrac{\partial f(x,y)}{\partial y} \end{bmatrix} \tag{10-16}$$

This vector has the well-known property that it points in the direction of maximum rate of change of f at (x, y) (see Problem 10.10). Equation (10-16) is valid at an arbitrary (but *single*) point (x, y). When evaluated for all applicable values of x and y, $\nabla f(x, y)$ becomes a *vector image*, each element of which is a vector given by Eq. (10-16). The *magnitude*, $M(x, y)$, of this gradient vector at a point (x, y) is given by its Euclidean vector norm:

$$M(x,y) = \|\nabla f(x,y)\| = \sqrt{g_x^2(x,y) + g_y^2(x,y)} \tag{10-17}$$

This is the *value* of the rate of change in the direction of the gradient vector at point (x, y). Note that $M(x, y)$, $\|\nabla f(x, y)\|$, $g_x(x, y)$, and $g_y(x, y)$ are arrays of the same size as f, created when x and y are allowed to vary over all pixel locations in f. It is common practice to refer to $M(x, y)$ and $\|\nabla f(x, y)\|$ as the *gradient image*, or simply as the *gradient* when the meaning is clear. The summation, square, and square root operations are elementwise operations, as defined in Section 2.6.

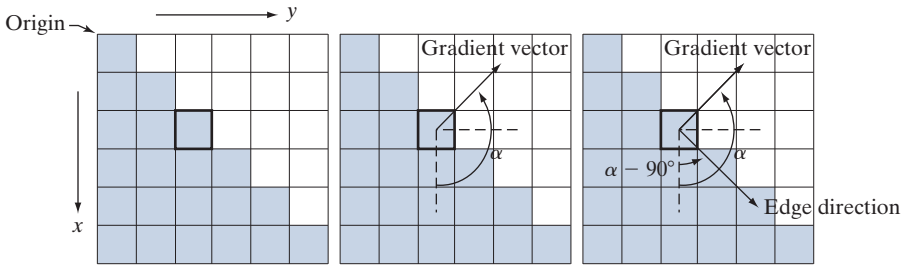
The *direction* of the gradient vector at a point (x, y) is given by

$$\alpha(x,y) = \tan^{-1}\left[\frac{g_y(x,y)}{g_x(x,y)}\right] \tag{10-18}$$

Angles are measured in the counterclockwise direction with respect to the x-axis (see Fig. 2.19). This is also an image of the same size as f, created by the elementwise division of g_x and g_y over all applicable values of x and y. The following example illustrates, the direction of an edge at a point (x, y) is orthogonal to the direction, $\alpha(x, y)$, of the gradient vector at the point.

EXAMPLE 10.5: Computing the gradient.

Figure 10.12(a) shows a zoomed section of an image containing a straight edge segment. Each square corresponds to a pixel, and we are interested in obtaining the strength and direction of the edge at the point highlighted with a box. The shaded pixels in this figure are assumed to have value 0, and the white



a b c

FIGURE 10.12 Using the gradient to determine edge strength and direction at a point. Note that the edge direction is perpendicular to the direction of the gradient vector at the point where the gradient is computed. Each square represents one pixel. (Recall from Fig. 2.19 that the origin of our coordinate system is at the top, left.)

pixels have value 1. We discuss after this example an approach for computing the derivatives in the x- and y-directions using a 3×3 neighborhood centered at a point. The method consists of subtracting the pixels in the top row of the neighborhood from the pixels in the bottom row to obtain the partial derivative in the x-direction. Similarly, we subtract the pixels in the left column from the pixels in the right column of the neighborhood to obtain the partial derivative in the y-direction. It then follows, using these differences as our estimates of the partials, that $\partial f/\partial x = -2$ and $\partial f/\partial y = 2$ at the point in question. Then,

$$\nabla f = \begin{bmatrix} g_x \\ g_y \end{bmatrix} = \begin{bmatrix} \dfrac{\partial f}{\partial x} \\ \dfrac{\partial f}{\partial y} \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$$

from which we obtain $\|\nabla f\| = 2\sqrt{2}$ at that point. Similarly, the direction of the gradient vector at the same point follows from Eq. (10-18): $\alpha = \tan^{-1}\left(g_y/g_x\right) = -45°$, which is the same as 135° measured in the positive (counterclockwise) direction with respect to the x-axis in our image coordinate system (see Fig. 2.19). Figure 10.12(b) shows the gradient vector and its direction angle.

As mentioned earlier, the direction of an edge at a point is orthogonal to the gradient vector at that point. So the direction angle of the edge in this example is $\alpha - 90° = 135° - 90° = 45°$, as Fig. 10.12(c) shows. All edge points in Fig. 10.12(a) have the same gradient, so the entire edge segment is in the same direction. The gradient vector sometimes is called the *edge normal*. When the vector is normalized to unit length by dividing it by its magnitude, the resulting vector is referred to as the *edge unit normal*.

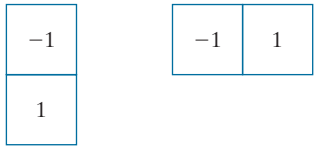
Gradient Operators

Obtaining the gradient of an image requires computing the partial derivatives $\partial f/\partial x$ and $\partial f/\partial y$ at every pixel location in the image. For the gradient, we typically use a forward or centered finite difference (see Table 10.1). Using forward differences we obtain

$$g_x(x,y) = \frac{\partial f(x,y)}{\partial x} = f(x+1,y) - f(x,y) \tag{10-19}$$

a b

FIGURE 10.13
1-D kernels used to implement Eqs. (10-19) and (10-20).



and

$$g_y(x,y) = \frac{\partial f(x,y)}{\partial y} = f(x,y+1) - f(x,y) \tag{10-20}$$

These two equations can be implemented for all values of x and y by filtering $f(x,y)$ with the 1-D kernels in Fig. 10.13.

When diagonal edge direction is of interest, we need 2-D kernels. The *Roberts cross-gradient operators* (Roberts [1965]) are one of the earliest attempts to use 2-D kernels with a diagonal preference. Consider the 3×3 region in Fig. 10.14(a). The Roberts operators are based on implementing the diagonal differences

Filter kernels used to compute the derivatives needed for the gradient are often called *gradient operators*, *difference operators*, *edge operators*, or *edge detectors*.

$$g_x = \frac{\partial f}{\partial x} = (z_9 - z_5) \tag{10-21}$$

and

$$g_y = \frac{\partial f}{\partial y} = (z_8 - z_6) \tag{10-22}$$

These derivatives can be implemented by filtering an image with the kernels shown in Figs. 10.14(b) and (c).

Kernels of size 2×2 are simple conceptually, but they are not as useful for computing edge direction as kernels that are symmetric about their centers, the smallest of which are of size 3×3. These kernels take into account the nature of the data on opposite sides of the center point, and thus carry more information regarding the direction of an edge. The simplest digital approximations to the partial derivatives using kernels of size 3×3 are given by

Observe that these two equations are first-order central differences as given in Eq. (10-6), but multiplied by 2.

$$g_x = \frac{\partial f}{\partial x} = (z_7 + z_8 + z_9) - (z_1 + z_2 + z_3)$$

and (10-23)

$$g_y = \frac{\partial f}{\partial y} = (z_3 + z_6 + z_9) - (z_1 + z_4 + z_7)$$

In this formulation, the difference between the third and first rows of the 3×3 region approximates the derivative in the x-direction, and the difference between the third and first columns approximate the derivative in the y-direction. Intuitively, we would expect these approximations to be more accurate than the approximations obtained using the Roberts operators. Equations (10-22) and (10-23) can be implemented over an entire image by filtering it with the two kernels in Figs. 10.14(d) and (e). These kernels are called the *Prewitt operators* (Prewitt [1970]).

A slight variation of the preceding two equations uses a weight of 2 in the center coefficient:

a
b c
d e
f g

FIGURE 10.14
A 3×3 region of an image (the z's are intensity values), and various kernels used to compute the gradient at the point labeled z_5.

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

−1	0	0	−1
0	1	1	0

Roberts

−1	−1	−1	−1	0	1
0	0	0	−1	0	1
1	1	1	−1	0	1

Prewitt

−1	−2	−1	−1	0	1
0	0	0	−2	0	2
1	2	1	−1	0	1

Sobel

$$g_x = \frac{\partial f}{\partial x} = (z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3) \tag{10-24}$$

and

$$g_y = \frac{\partial f}{\partial y} = (z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7) \tag{10-25}$$

It can be demonstrated (see Problem 10.12) that using a 2 in the center location provides image smoothing. Figures 10.14(f) and (g) show the kernels used to implement Eqs. (10-24) and (10-25). These kernels are called the *Sobel operators* (Sobel [1970]).

The Prewitt kernels are simpler to implement than the Sobel kernels, but the slight computational difference between them typically is not an issue. The fact that the Sobel kernels have better noise-suppression (smoothing) characteristics makes them preferable because, as mentioned earlier in the discussion of Fig. 10.11, noise suppression is an important issue when dealing with derivatives. Note that the

a b

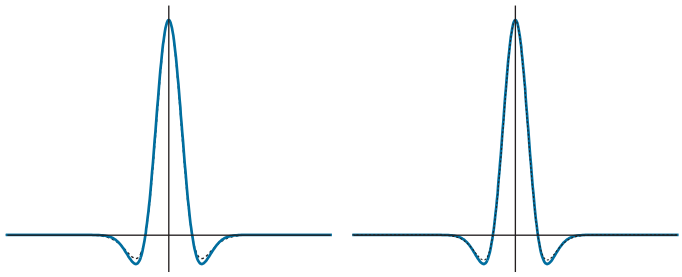


FIGURE 10.23
(a) Negatives of the LoG (solid) and DoG (dotted) profiles using a σ ratio of 1.75:1. (b) Profiles obtained using a ratio of 1.6:1.

Gaussian kernels are separable (see Section 3.4). Therefore, both the LoG and the DoG filtering operations can be implemented with 1-D convolutions instead of using 2-D convolutions directly (see Problem 10.19). For an image of size $M \times N$ and a kernel of size $n \times n$, doing so reduces the number of multiplications and additions for each convolution from being proportional to n^2MN for 2-D convolutions to being proportional to nMN for 1-D convolutions. This implementation difference is significant. For example, if $n = 25$, a 1-D implementation will require on the order of 12 times fewer multiplication and addition operations than using 2-D convolution.

The Canny Edge Detector

Although the algorithm is more complex, the performance of the Canny edge detector (Canny [1986]) discussed in this section is superior in general to the edge detectors discussed thus far. Canny's approach is based on three basic objectives:

1. *Low error rate*. All edges should be found, and there should be no spurious responses.
2. *Edge points should be well localized*. The edges located must be as close as possible to the true edges. That is, the distance between a point marked as an edge by the detector and the center of the true edge should be minimum.
3. *Single edge point response*. The detector should return only one point for each true edge point. That is, the number of local maxima around the true edge should be minimum. This means that the detector should not identify multiple edge pixels where only a single edge point exists.

The essence of Canny's work was in expressing the preceding three criteria mathematically, and then attempting to find optimal solutions to these formulations. In general, it is difficult (or impossible) to find a closed-form solution that satisfies all the preceding objectives. However, using numerical optimization with 1-D step edges corrupted by additive white Gaussian noise[†] led to the conclusion that a good approximation to the optimal step edge detector is the *first derivative of a Gaussian*,

$$\frac{d}{dx}e^{-\frac{x^2}{2\sigma^2}} = \frac{-x}{\sigma^2}e^{-\frac{x^2}{2\sigma^2}} \tag{10-34}$$

[†]Recall that *white noise* is noise having a frequency spectrum that is continuous and uniform over a specified frequency band. *White Gaussian noise* is white noise in which the distribution of amplitude values is Gaussian. Gaussian white noise is a good approximation of many real-world situations and generates mathematically tractable models. It has the useful property that its values are statistically independent.

where the approximation was only about 20% worse that using the optimized numerical solution (a difference of this magnitude generally is visually imperceptible in most applications).

Generalizing the preceding result to 2-D involves recognizing that the 1-D approach still applies in the direction of the edge normal (see Fig. 10.12). Because the direction of the normal is unknown beforehand, this would require applying the 1-D edge detector in all possible directions. This task can be approximated by first smoothing the image with a circular 2-D Gaussian function, computing the gradient of the result, and then using the gradient magnitude and direction to estimate edge strength and direction at every point.

Let $f(x, y)$ denote the input image and $G(x, y)$ denote the Gaussian function:

$$G(x, y) = e^{-\frac{x^2 + y^2}{2\sigma^2}} \tag{10-35}$$

We form a smoothed image, $f_s(x, y)$, by convolving f and G:

$$f_s(x, y) = G(x, y) \star f(x, y) \tag{10-36}$$

This operation is followed by computing the gradient magnitude and direction (angle), as discussed earlier:

$$M_s(x, y) = \|\nabla f_s(x, y)\| = \sqrt{g_x^2(x, y) + g_y^2(x, y)} \tag{10-37}$$

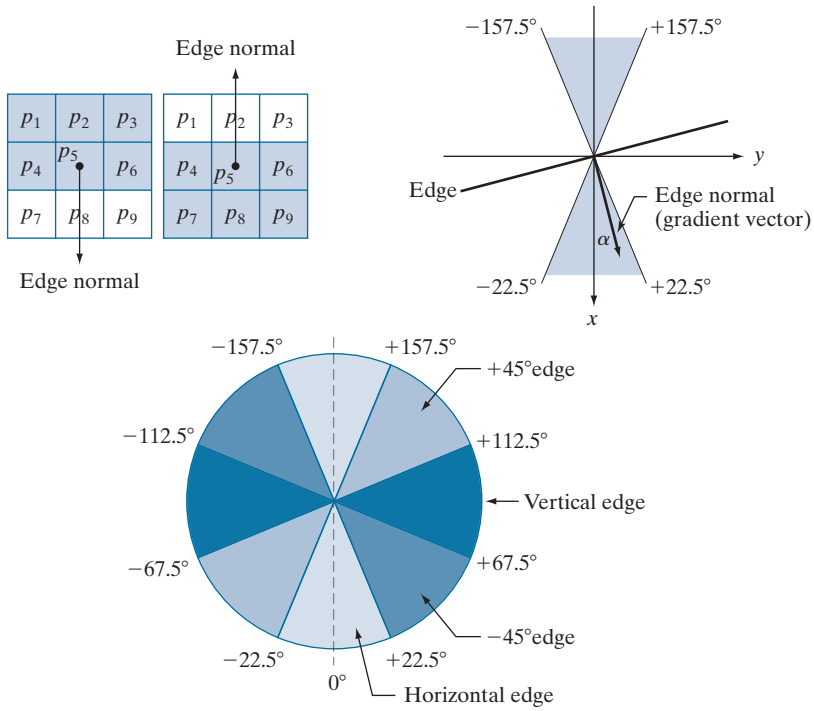
and

$$\alpha(x, y) = \tan^{-1}\left[\frac{g_y(x, y)}{g_x(x, y)}\right] \tag{10-38}$$

with $g_x(x, y) = \partial f_s(x, y)/\partial x$ and $g_y(x, y) = \partial f_s(x, y)/\partial y$. Any of the derivative filter kernel pairs in Fig. 10.14 can be used to obtain $g_x(x, y)$ and $g_y(x, y)$. Equation (10-36) is implemented using an $n \times n$ Gaussian kernel whose size is discussed below. Keep in mind that $\|\nabla f_s(x, y)\|$ and $\alpha(x, y)$ are arrays of the same size as the image from which they are computed.

Gradient image $\|\nabla f_s(x, y)\|$ typically contains wide ridges around local maxima. The next step is to thin those ridges. One approach is to use *nonmaxima suppression*. The essence of this approach is to specify a number of discrete orientations of the edge normal (gradient vector). For example, in a 3×3 region we can define four orientations[†] for an edge passing through the center point of the region: horizontal, vertical, +45°, and −45°. Figure 10.24(a) shows the situation for the two possible orientations of a horizontal edge. Because we have to quantize all possible edge directions into four ranges, we have to define a range of directions over which we consider an edge to be horizontal. We determine edge direction from the direction of the edge normal, which we obtain directly from the image data using Eq. (10-38). As Fig. 10.24(b) shows, if the edge normal is in the range of directions from −22.5° to

[†]Every edge has two possible orientations. For example, an edge whose normal is oriented at 0° and an edge whose normal is oriented at 180° are the same *horizontal* edge.



a b
c

FIGURE 10.24
(a) Two possible orientations of a horizontal edge (shaded) in a 3×3 neighborhood. (b) Range of values (shaded) of α, the direction angle of the edge normal for a horizontal edge. (c) The angle ranges of the edge normals for the four types of edge directions in a 3×3 neighborhood. Each edge direction has two ranges, shown in corresponding shades.

22.5° or from −157.5° to 157.5°, we call the edge a horizontal edge. Figure 10.24(c) shows the angle ranges corresponding to the four directions under consideration.

Let d_1, d_2, d_3,and d_4 denote the four basic edge directions just discussed for a 3×3 region: horizontal, −45°, vertical, and +45°, respectively. We can formulate the following nonmaxima suppression scheme for a 3×3 region centered at an arbitrary point (x, y) in α:

1. Find the direction d_k that is closest to $\alpha(x, y)$.
2. Let K denote the value of $\|\nabla f_s\|$ at (x, y). If K is less than the value of $\|\nabla f_s\|$ at one or both of the neighbors of point (x, y) along d_k, let $g_N(x, y) = 0$ (suppression); otherwise, let $g_N(x, y) = K$.

When repeated for all values of x and y, this procedure yields a nonmaxima suppressed image $g_N(x, y)$ that is of the same size as $f_s(x, y)$. For example, with reference to Fig. 10.24(a), letting (x, y) be at p_5, and assuming a horizontal edge through p_5, the pixels of interest in Step 2 would be p_2 and p_8. Image $g_N(x, y)$ contains only the thinned edges; it is equal to image $\|\nabla f_s(x, y)\|$ with the nonmaxima edge points suppressed.

The final operation is to threshold $g_N(x, y)$ to reduce false edge points. In the Marr-Hildreth algorithm we did this using a single threshold, in which all values below the threshold were set to 0. If we set the threshold too low, there will still be some false edges (called *false positives*). If the threshold is set too high, then valid edge points will be eliminated (*false negatives*). Canny's algorithm attempts to

improve on this situation by using *hysteresis thresholding* which, as we will discuss in Section 10.3, uses two thresholds: a low threshold, T_L and a high threshold, T_H. Experimental evidence (Canny [1986]) suggests that the ratio of the high to low threshold should be in the range of 2:1 to 3:1.

We can visualize the thresholding operation as creating two additional images:

$$g_{NH}(x,y) = g_N(x,y) \geq T_H \tag{10-39}$$

and

$$g_{NL}(x,y) = g_N(x,y) \geq T_L \tag{10-40}$$

Initially, $g_{NH}(x,y)$ and $g_{NL}(x,y)$ are set to 0. After thresholding, $g_{NH}(x,y)$ will usually have fewer nonzero pixels than $g_{NL}(x,y)$, but all the nonzero pixels in $g_{NH}(x,y)$ will be contained in $g_{NL}(x,y)$ because the latter image is formed with a lower threshold. We eliminate from $g_{NL}(x,y)$ all the nonzero pixels from $g_{NH}(x,y)$ by letting

$$g_{NL}(x,y) = g_{NL}(x,y) - g_{NH}(x,y) \tag{10-41}$$

The nonzero pixels in $g_{NH}(x,y)$ and $g_{NL}(x,y)$ may be viewed as being "strong" and "weak" edge pixels, respectively. After the thresholding operations, all strong pixels in $g_{NH}(x,y)$ are assumed to be valid edge pixels, and are so marked immediately. Depending on the value of T_H, the edges in $g_{NH}(x,y)$ typically have gaps. Longer edges are formed using the following procedure:

(a) Locate the next unvisited edge pixel, p, in $g_{NH}(x,y)$.

(b) Mark as valid edge pixels all the weak pixels in $g_{NL}(x,y)$ that are connected to p using, say, 8-connectivity.

(c) If all nonzero pixels in $g_{NH}(x,y)$ have been visited go to Step (d). Else, return to Step (a).

(d) Set to zero all pixels in $g_{NL}(x,y)$ that were not marked as valid edge pixels.

At the end of this procedure, the final image output by the Canny algorithm is formed by appending to $g_{NH}(x,y)$ all the nonzero pixels from $g_{NL}(x,y)$.

We used two additional images, $g_{NH}(x,y)$ and $g_{NL}(x,y)$ to simplify the discussion. In practice, hysteresis thresholding can be implemented directly during nonmaxima suppression, and thresholding can be implemented directly on $g_N(x,y)$ by forming a list of strong pixels and the weak pixels connected to them.

Summarizing, the Canny edge detection algorithm consists of the following steps:

1. Smooth the input image with a Gaussian filter.

2. Compute the gradient magnitude and angle images.

3. Apply nonmaxima suppression to the gradient magnitude image.

4. Use double thresholding and connectivity analysis to detect and link edges.

Although the edges after nonmaxima suppression are thinner than raw gradient edges, the former can still be thicker than one pixel. To obtain edges one pixel thick, it is typical to follow Step 4 with one pass of an edge-thinning algorithm (see Section 9.5).

In terms of edge quality and the ability to eliminate irrelevant detail, the results in Fig. 10.26 correspond closely to the results and conclusions in the previous example. Note also that the Canny algorithm was the only procedure capable of yielding a totally unbroken edge for the posterior boundary of the brain, and the closest boundary of the spinal cord. It was also the only procedure capable of finding the cleanest contours, while eliminating all the edges associated with the gray brain matter in the original image.

The price paid for the improved performance of the Canny algorithm is a significantly more complex implementation than the two approaches discussed earlier. In some applications, such as real-time industrial image processing, cost and speed requirements usually dictate the use of simpler techniques, principally the thresholded gradient approach. When edge quality is the driving force, the Marr-Hildreth and Canny algorithms, especially the latter, offer superior alternatives.

LINKING EDGE POINTS

Ideally, edge detection should yield sets of pixels lying only on edges. In practice, these pixels seldom characterize edges completely because of noise, breaks in the edges caused by nonuniform illumination, and other effects that introduce discontinuities in intensity values. Therefore, edge detection typically is followed by linking algorithms designed to assemble edge pixels into meaningful edges and/or region boundaries. In this section, we discuss two fundamental approaches to edge linking that are representative of techniques used in practice. The first requires knowledge about edge points in a local region (e.g., a 3×3 neighborhood), and the second is a global approach that works with an entire edge map. As it turns out, linking points along the boundary of a region is also an important aspect of some of the segmentation methods discussed in the next chapter, and in extracting features from a segmented image, as we will do in Chapter 11. Thus, you will encounter additional edge-point linking methods in the next two chapters.

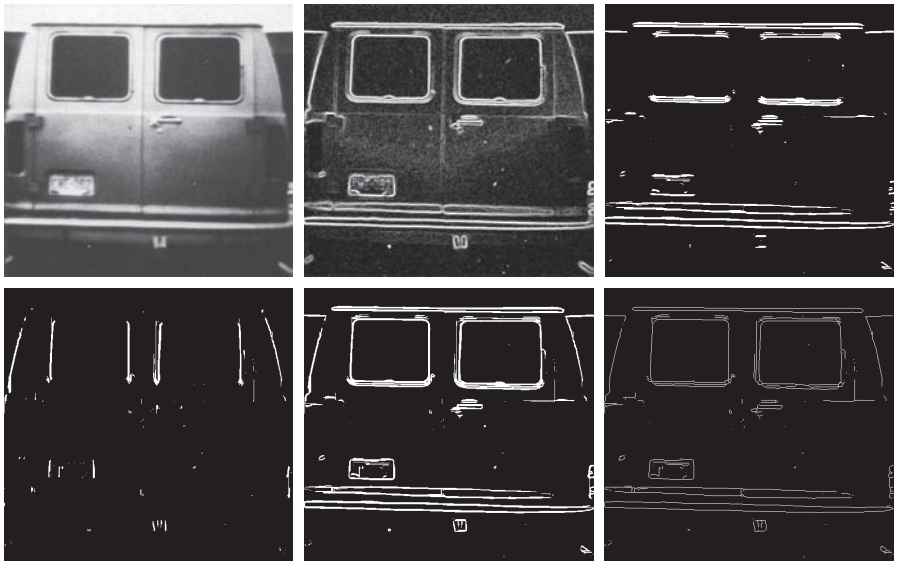
Local Processing

A simple approach for linking edge points is to analyze the characteristics of pixels in a small neighborhood about every point (x, y) that has been declared an edge point by one of the techniques discussed in the preceding sections. All points that are similar according to predefined criteria are linked, forming an edge of pixels that share common properties according to the specified criteria.

The two principal properties used for establishing similarity of edge pixels in this kind of local analysis are (1) the strength (magnitude) and (2) the direction of the gradient vector. The first property is based on Eq. (10-17). Let S_{xy} denote the set of coordinates of a neighborhood centered at point (x, y) in an image. An edge pixel with coordinates (s,t) in S_{xy} is similar in *magnitude* to the pixel at (x, y) if

$$|M(s,t) - M(x,y)| \leq E \tag{10-42}$$

where E is a positive threshold.



a b c
d e f

FIGURE 10.27
(a) Image of the rear of a vehicle. (b) Gradient magnitude image. (c) Horizontally connected edge pixels. (d) Vertically connected edge pixels. (e) The logical OR of (c) and (d). (f) Final result, using morphological thinning. (Original image courtesy of Perceptics Corporation.)

strong horizontal and vertical edges. Figure 10.27(b) shows the gradient magnitude image, $M(x, y)$, and Figs. 10.27(c) and (d) show the result of Steps 3 and 4 of the algorithm, obtained by letting T_M equal to 30% of the maximum gradient value, $A = 90°$, $T_A = 45°$, and filling all gaps of 25 or fewer pixels (approximately 5% of the image width). A large range of allowable angle directions was required to detect the rounded corners of the license plate enclosure, as well as the rear windows of the vehicle. Figure 10.27(e) is the result of forming the logical OR of the two preceding images, and Fig. 10.27(f) was obtained by thinning 10.27(e) with the thinning procedure discussed in Section 9.5. As Fig. 10.27(f) shows, the rectangle corresponding to the license plate was clearly detected in the image. It would be a simple matter to isolate the license plate from all the rectangles in the image, using the fact that the width-to-height ratio of license plates have distinctive proportions (e.g., a 2:1 ratio in U.S. plates).

Global Processing Using the Hough Transform

The method discussed in the previous section is applicable in situations in which knowledge about pixels belonging to individual objects is available. Often, we have to work in unstructured environments in which all we have is an edge map and no knowledge about where objects of interest might be. In such situations, all pixels are candidates for linking, and thus have to be accepted or eliminated based on predefined *global* properties. In this section, we develop an approach based on whether sets of pixels lie on curves of a specified shape. Once detected, these curves form the edges or region boundaries of interest.

Given n points in an image, suppose that we want to find subsets of these points that lie on straight lines. One possible solution is to find all lines determined by every pair of points, then find all subsets of points that are close to particular lines. This approach involves finding $n(n-1)/2 \sim n^2$ lines, then performing $(n)(n(n-1))/2 \sim n^3$

comparisons of every point to all lines. This is a computationally prohibitive task in most applications.

The original formulation of the Hough transform presented here works with straight lines. For a generalization to arbitrary shapes, see Ballard [1981].

Hough [1962] proposed an alternative approach, commonly referred to as the *Hough transform*. Let (x_i, y_i) denote a point in the xy-plane and consider the general equation of a straight line in slope-intercept form: $y_i = ax_i + b$. Infinitely many lines pass through (x_i, y_i), but they all satisfy the equation $y_i = ax_i + b$ for varying values of a and b. However, writing this equation as $b = -x_i a + y_i$ and considering the ab-plane (also called *parameter space*) yields the equation of a *single* line for a fixed point (x_i, y_i). Furthermore, a second point (x_j, y_j) also has a single line in parameter space associated with it, which intersects the line associated with (x_i, y_i) at some point (a', b') in parameter space, where a' is the slope and b' the intercept of the line containing both (x_i, y_i) and (x_j, y_j) in the xy-plane (we are assuming, of course, that the lines are not parallel). In fact, *all* points on this line have lines in parameter space that intersect at (a', b'). Figure 10.28 illustrates these concepts.

In principle, the parameter space lines corresponding to all points (x_k, y_k) in the xy-plane could be plotted, and the principal lines in that plane could be found by identifying points in parameter space where large numbers of parameter-space lines intersect. However, a difficulty with this approach is that a, (the slope of a line) approaches infinity as the line approaches the vertical direction. One way around this difficulty is to use *the normal representation* of a line:

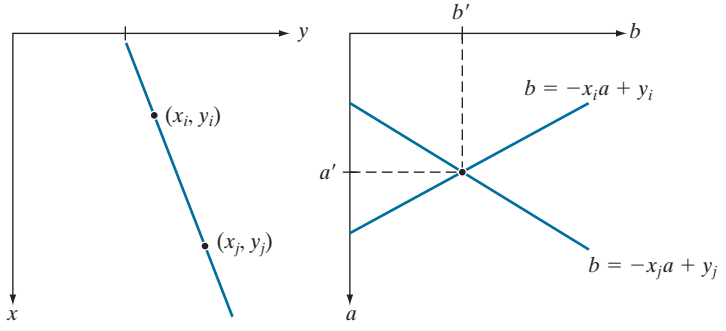
$$x\cos\theta + y\sin\theta = \rho \tag{10-44}$$

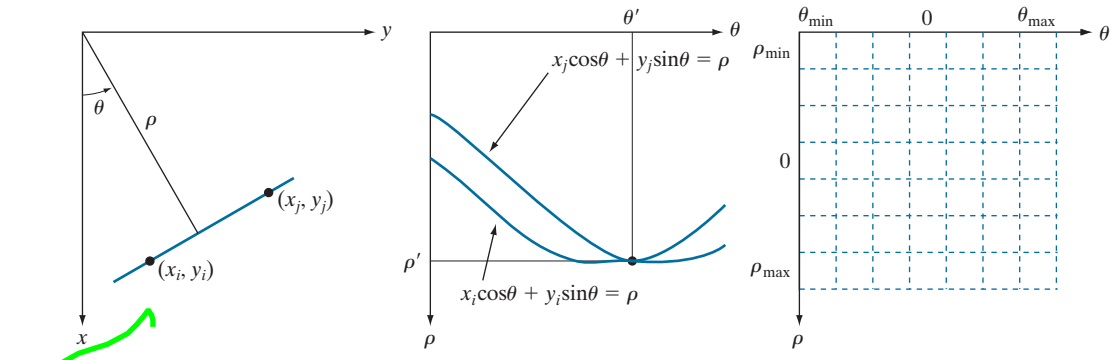
Figure 10.29(a) illustrates the geometrical interpretation of the parameters ρ and θ. A horizontal line has $\theta = 0°$, with ρ being equal to the positive x-intercept. Similarly, a vertical line has $\theta = 90°$, with ρ being equal to the positive y-intercept, or $\theta = -90°$, with ρ being equal to the negative y-intercept (we limit the angle to the range $-90° \le \theta \le 90°$). Each sinusoidal curve in Figure 10.29(b) represents the family of lines that pass through a particular point (x_k, y_k) in the xy-plane. The intersection point (ρ', θ') in Fig. 10.29(b) corresponds to the line that passes through both (x_i, y_i) and (x_j, y_j) in Fig. 10.29(a).

The computational attractiveness of the Hough transform arises from subdividing the $\rho\theta$ parameter space into so-called *accumulator cells*, as Fig. 10.29(c) illustrates, where $(\rho_{\min}, \rho_{\max})$ and $(\theta_{\min}, \theta_{\max})$ are the expected ranges of the parameter values:

a b

FIGURE 10.28
(a) xy-plane. (b) Parameter space.





a b c

FIGURE 10.29 (a) (ρ,θ) parameterization of a line in the xy-plane. (b) Sinusoidal curves in the $\rho\theta$-plane; the point of intersection (ρ',θ') corresponds to the line passing through points (x_i,y_i) and (x_j,y_j) in the xy-plane. (c) Division of the $\rho\theta$-plane into accumulator cells.

$-90° \le \theta \le 90°$ and $-D \le \rho \le D$, where D is the maximum distance between opposite corners in an image. The cell at coordinates (i,j) with accumulator value $A(i,j)$ corresponds to the square associated with parameter-space coordinates (ρ_i,θ_j). Initially, these cells are set to zero. Then, for every non-background point (x_k,y_k) in the xy-plane, we let θ equal each of the allowed subdivision values on the θ-axis and solve for the corresponding ρ using the equation $\rho = x_k \cos\theta + y_k \sin\theta$. The resulting ρ values are then rounded off to the nearest allowed cell value along the ρ axis. If a choice of θ_q results in the solution ρ_p, then we let $A(p,q) = A(p,q) + 1$. At the end of the procedure, a value of K in a cell $A(i,j)$ means that K points in the xy-plane lie on the line $x\cos\theta_j + y\sin\theta_j = \rho_i$. The number of subdivisions in the $\rho\theta$-plane determines the accuracy of the colinearity of these points. It can be shown (see Problem 10.27) that the number of computations in the method just discussed is linear with respect to n, the number of non-background points in the xy-plane.

EXAMPLE 10.11: Some basic properties of the Hough transform.

Figure 10.30 illustrates the Hough transform based on Eq. (10-44). Figure 10.30(a) shows an image of size $M \times M$ ($M = 101$) with five labeled white points, and Fig. 10.30(b) shows each of these points mapped onto the $\rho\theta$-plane using subdivisions of one unit for the ρ and θ axes. The range of θ values is $\pm 90°$, and the range of ρ values is $\pm\sqrt{2}M$. As Fig. 10.30(b) shows, each curve has a different sinusoidal shape. The horizontal line resulting from the mapping of point 1 is a sinusoid of zero amplitude.

The points labeled A (not to be confused with accumulator values) and B in Fig. 10.30(b) illustrate the colinearity detection property of the Hough transform. For example, point B, marks the intersection of the curves corresponding to points 2, 3, and 4 in the xy image plane. The location of point A indicates that these three points lie on a straight line passing through the origin ($\rho = 0$) and oriented at $-45°$ [see Fig. 10.29(a)]. Similarly, the curves intersecting at point B in parameter space indicate that points 2, 3, and 4 lie on a straight line oriented at 45°, and whose distance from the origin is $\rho = 71$ (one-half the diagonal distance from the origin of the image to the opposite corner, rounded to the nearest integer

a
b

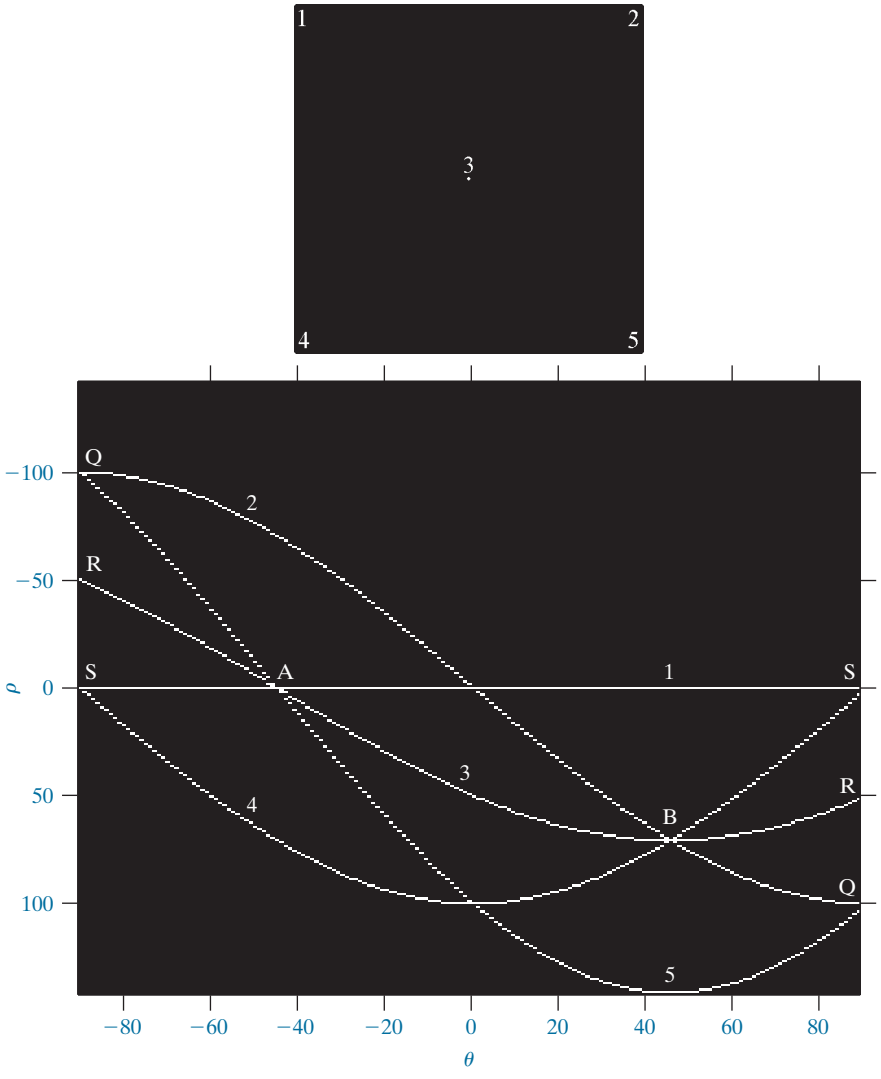


FIGURE 10.30
(a) Image of size 101×101 pixels, containing five white points (four in the corners and one in the center). (b) Corresponding parameter space.

value). Finally, the points labeled Q, R, and S in Fig. 10.30(b) illustrate the fact that the Hough transform exhibits a reflective adjacency relationship at the right and left edges of the parameter space. This property is the result of the manner in which ρ and θ change sign at the $\pm 90°$ boundaries.

Although the focus thus far has been on straight lines, the Hough transform is applicable to any function of the form $g(\mathbf{v}, \mathbf{c}) = 0$, where $\mathbf{v}$ is a vector of coordinates and $\mathbf{c}$ is a vector of coefficients. For example, points lying on the circle

$$(x - c_1)^2 + (y - c_2)^2 = c_3^2 \tag{10-45}$$

can be detected by using the basic approach just discussed. The difference is the presence of three parameters c_1, c_2, and c_3 that result in a 3-D parameter space with

cube-like cells, and accumulators of the form $A(i, j, k)$. The procedure is to increment c_1 and c_2, solve for the value of c_3 that satisfies Eq. (10-45), and update the accumulator cell associated with the triplet (c_1, c_2, c_3). Clearly, the complexity of the Hough transform depends on the number of coordinates and coefficients in a given functional representation. As noted earlier, generalizations of the Hough transform to detect curves with no simple analytic representations are possible, as is the application of the transform to grayscale images.

Returning to the edge-linking problem, an approach based on the Hough transform is as follows:

1. Obtain a binary edge map using any of the methods discussed earlier in this section.
2. Specify subdivisions in the $\rho\theta$-plane.
3. Examine the counts of the accumulator cells for high pixel concentrations.
4. Examine the relationship (principally for continuity) between pixels in a chosen cell.

Continuity in this case usually is based on computing the distance between disconnected pixels corresponding to a given accumulator cell. A gap in a line associated with a given cell is bridged if the length of the gap is less than a specified threshold. Being able to group lines based on direction is a global concept applicable over the entire image, requiring only that we examine pixels associated with specific accumulator cells. The following example illustrates these concepts.

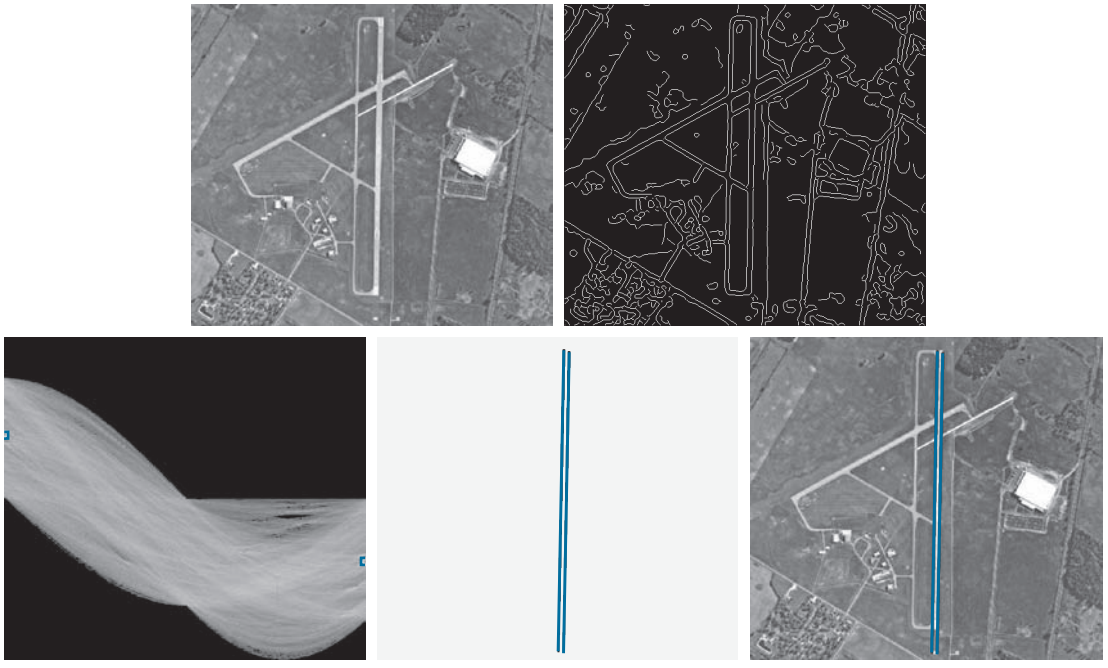
EXAMPLE 10.12: Using the Hough transform for edge linking.

Figure 10.31(a) shows an aerial image of an airport. The objective of this example is to use the Hough transform to extract the two edges defining the principal runway. A solution to such a problem might be of interest, for instance, in applications involving autonomous air navigation.

The first step is to obtain an edge map. Figure 10.31(b) shows the edge map obtained using Canny's algorithm with the same parameters and procedure used in Example 10.9. For the purpose of computing the Hough transform, similar results can be obtained using any of the other edge-detection techniques discussed earlier. Figure 10.31(c) shows the Hough parameter space obtained using 1° increments for θ, and one-pixel increments for ρ.

The runway of interest is oriented approximately 1° off the north direction, so we select the cells corresponding to ±90° and containing the highest count because the runways are the longest lines oriented in these directions. The small boxes on the edges of Fig. 10.31(c) highlight these cells. As mentioned earlier in connection with Fig. 10.30(b), the Hough transform exhibits adjacency at the edges. Another way of interpreting this property is that a line oriented at +90° and a line oriented at −90° are equivalent (i.e., they are both vertical). Figure 10.31(d) shows the lines corresponding to the two accumulator cells just discussed, and Fig. 10.31(e) shows the lines superimposed on the original image. The lines were obtained by joining all gaps not exceeding 20% (approximately 100 pixels) of the image height. These lines clearly correspond to the edges of the runway of interest.

Note that the only information needed to solve this problem was the orientation of the runway and the observer's position relative to it. In other words, a vehicle navigating autonomously would know that if the runway of interest faces north, and the vehicle's direction of travel also is north, the runway should appear vertically in the image. Other relative orientations are handled in a similar manner. The



a b
c d e

FIGURE 10.31 (a) A 502×564 aerial image of an airport. (b) Edge map obtained using Canny's algorithm. (c) Hough parameter space (the boxes highlight the points associated with long vertical lines). (d) Lines in the image plane corresponding to the points highlighted by the boxes. (e) Lines superimposed on the original image.

orientations of runways throughout the world are available in flight charts, and the direction of travel is easily obtainable using GPS (Global Positioning System) information. This information also could be used to compute the distance between the vehicle and the runway, thus allowing estimates of parameters such as expected length of lines relative to image size, as we did in this example.

10.3 THRESHOLDING

Because of its intuitive properties, simplicity of implementation, and computational speed, image thresholding enjoys a central position in applications of image segmentation. Thresholding was introduced in Section 3.1, and we have used it in various discussions since then. In this section, we discuss thresholding in a more formal way, and develop techniques that are considerably more general than what has been presented thus far.

FOUNDATION

In the previous section, regions were identified by first finding edge segments, then attempting to link the segments into boundaries. In this section, we discuss

techniques for partitioning images directly into regions based on intensity values and/or properties of these values.

The Basics of Intensity Thresholding

Suppose that the intensity histogram in Fig. 10.32(a) corresponds to an image, $f(x, y)$, composed of light objects on a dark background, in such a way that object and background pixels have intensity values grouped into two dominant modes. One obvious way to extract the objects from the background is to select a threshold, T, that separates these modes. Then, any point (x, y) in the image at which $f(x, y) > T$ is called an *object point*. Otherwise, the point is called a *background* point. In other words, the segmented image, denoted by $g(x, y)$, is given by

Remember, $f(x, y)$ denotes the intensity of f at coordinates (x, y).

Although we follow convention in using 0 intensity for the background and 1 for object pixels, any two distinct values can be used in Eq. (10-46).

$$g(x, y) = \begin{cases} 1 & \text{if } f(x, y) > T \\ 0 & \text{if } f(x, y) \leq T \end{cases} \tag{10-46}$$

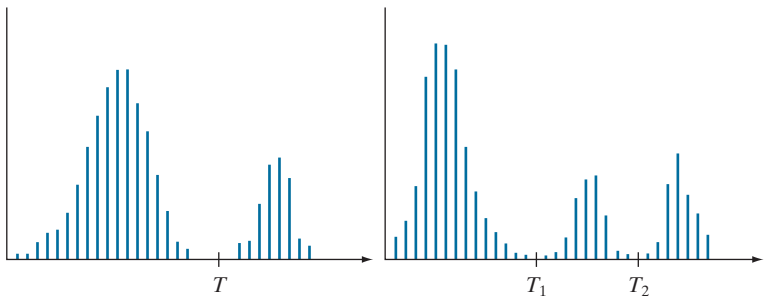
When T is a constant applicable over an entire image, the process given in this equation is referred to as *global thresholding*. When the value of T changes over an image, we use the term *variable thresholding*. The terms *local* or *regional* thresholding are used sometimes to denote variable thresholding in which the value of T at any point (x, y) in an image depends on properties of a neighborhood of (x, y) (for example, the average intensity of the pixels in the neighborhood). If T depends on the spatial coordinates (x, y) themselves, then variable thresholding is often referred to as *dynamic* or *adaptive* thresholding. Use of these terms is not universal.

Figure 10.32(b) shows a more difficult thresholding problem involving a histogram with three dominant modes corresponding, for example, to two types of light objects on a dark background. Here, *multiple thresholding* classifies a point (x, y) as belonging to the background if $f(x, y) \leq T_1$, to one object class if $T_1 < f(x, y) \leq T_2$, and to the other object class if $f(x, y) > T_2$. That is, the segmented image is given by

$$g(x, y) = \begin{cases} a & \text{if } f(x, y) > T_2 \\ b & \text{if } T_1 < f(x, y) \leq T_2 \\ c & \text{if } f(x, y) \leq T_1 \end{cases} \tag{10-47}$$

a b

FIGURE 10.32 Intensity histograms that can be partitioned (a) by a single threshold, and (b) by dual thresholds.



perfectly uniform, but the reflectance of the image was not, as a results, for example, of natural reflectivity variations in the surface of objects and/or background.

The important point is that illumination and reflectance play a central role in the success of image segmentation using thresholding or other segmentation techniques. Therefore, controlling these factors when possible should be the first step considered in the solution of a segmentation problem. There are three basic approaches to the problem when control over these factors is not possible. The first is to correct the shading pattern directly. For example, nonuniform (but fixed) illumination can be corrected by multiplying the image by the inverse of the pattern, which can be obtained by imaging a flat surface of constant intensity. The second is to attempt to correct the global shading pattern via processing using, for example, the top-hat transformation introduced in Section 9.8. The third approach is to "work around" nonuniformities using variable thresholding, as discussed later in this section.

BASIC GLOBAL THRESHOLDING

When the intensity distributions of objects and background pixels are sufficiently distinct, it is possible to use a single (*global*) threshold applicable over the entire image. In most applications, there is usually enough variability between images that, even if global thresholding is a suitable approach, an algorithm capable of estimating the threshold value for each image is required. The following iterative algorithm can be used for this purpose:

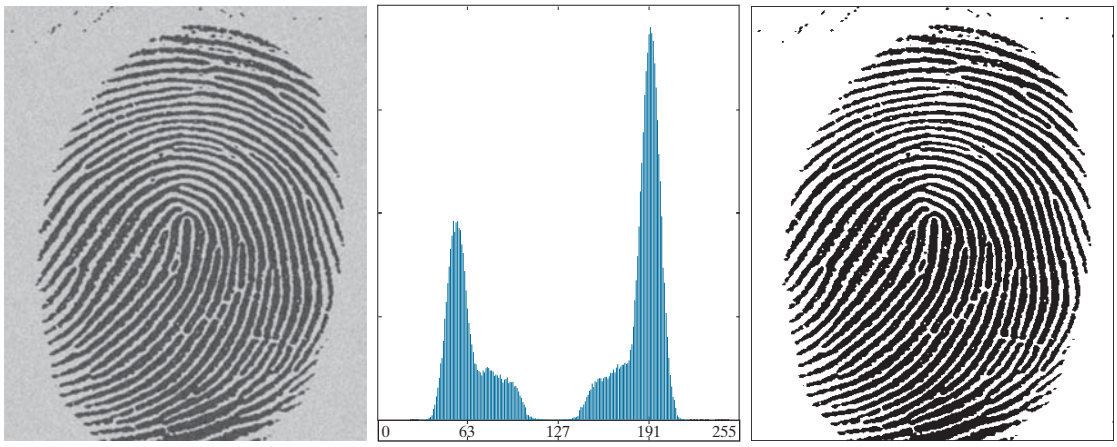
1. Select an initial estimate for the global threshold, T.
2. Segment the image using T in Eq. (10-46). This will produce two groups of pixels: G_1, consisting of pixels with intensity values $> T$; and G_2, consisting of pixels with values $\leq T$.
3. Compute the average (mean) intensity values m_1 and m_2 for the pixels in G_1 and G_2, respectively.
4. Compute a new threshold value midway between m_1 and m_2:

$$T = \frac{1}{2}(m_1 + m_2)$$

5. Repeat Steps 2 through 4 until the difference between values of T in successive iterations is smaller than a predefined value, ΔT.

The algorithm is stated here in terms of successively thresholding the input image and calculating the means at each step, because it is more intuitive to introduce it in this manner. However, it is possible to develop an equivalent (and more efficient) procedure by expressing all computations in the terms of the image histogram, which has to be computed only once (see Problem 10.29).

The preceding algorithm works well in situations where there is a reasonably clear valley between the modes of the histogram related to objects and background. Parameter ΔT is used to stop iterating when the changes in threshold values is small. The initial threshold must be chosen greater than the minimum and less than the maximum intensity level in the image (the average intensity of the image is a good



a b c

FIGURE 10.35 (a) Noisy fingerprint. (b) Histogram. (c) Segmented result using a global threshold (thin image border added for clarity). (Original image courtesy of the National Institute of Standards and Technology.).

initial choice for T). If this condition is met, the algorithm converges in a finite number of steps, whether or not the modes are separable (see Problem 10.30).

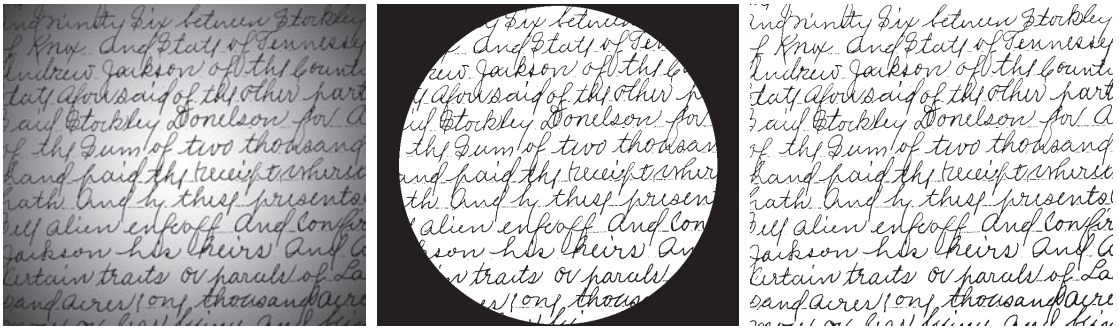
EXAMPLE 10.13: Global thresholding.

Figure 10.35 shows an example of segmentation using the preceding iterative algorithm. Figure 10.35(a) is the original image and Fig. 10.35(b) is the image histogram, showing a distinct valley. Application of the basic global algorithm resulted in the threshold $T = 125.4$ after three iterations, starting with T equal to the average intensity of the image, and using $\Delta T = 0$. Figure 10.35(c) shows the result obtained using $T = 125$ to segment the original image. As expected from the clear separation of modes in the histogram, the segmentation between object and background was perfect.

OPTIMUM GLOBAL THRESHOLDING USING OTSU'S METHOD

Thresholding may be viewed as a statistical-decision theory problem whose objective is to minimize the average error incurred in assigning pixels to two or more groups (also called *classes*). This problem is known to have an elegant closed-form solution known as the *Bayes decision function* (see Section 12.4). The solution is based on only two parameters: the probability density function (PDF) of the intensity levels of each class, and the probability that each class occurs in a given application. Unfortunately, estimating PDFs is not a trivial matter, so the problem usually is simplified by making workable assumptions about the form of the PDFs, such as assuming that they are Gaussian functions. Even with simplifications, the process of implementing solutions using these assumptions can be complex and not always well-suited for real-time applications.

The approach in the following discussion, called *Otsu's method* (Otsu [1979]), is an attractive alternative. The method is optimum in the sense that it maximizes the



a b c

FIGURE 10.44 (a) Text image corrupted by spot shading. (b) Result of global thresholding using Otsu's method. (c) Result of local thresholding using moving averages.

As another illustration of the effectiveness of this segmentation approach, we used the same parameters as in the previous paragraph to segment the image in Fig. 10.45(a), which is corrupted by a sinusoidal intensity variation typical of the variations that may occur when the power supply in a document scanner is not properly grounded. As Figs. 10.45(b) and (c) show, the segmentation results are comparable to those in Fig. 10.44.

Note that successful segmentation results were obtained in both cases using the same values for n and c, which shows the relative ruggedness of the approach. In general, thresholding based on moving averages works well when the objects of interest are small (or thin) with respect to the image size, a condition satisfied by images of typed or handwritten text.

10.4 SEGMENTATION BY REGION GROWING AND BY REGION SPLITTING AND MERGING

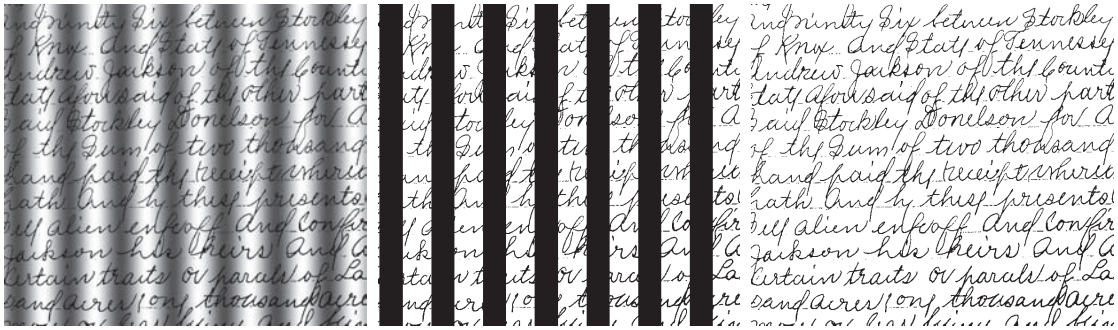
You should review the terminology introduced in Section 10.1 before proceeding.

As we discussed in Section 10.1, the objective of segmentation is to partition an image into regions. In Section 10.2, we approached this problem by attempting to find boundaries between regions based on discontinuities in intensity levels, whereas in Section 10.3, segmentation was accomplished via thresholds based on the distribution of pixel properties, such as intensity values or color. In this section and in Sections 10.5 and 10.6, we discuss segmentation techniques that find the regions directly. In Section 10.7, we will discuss a method that finds the regions and their boundaries simultaneously.

REGION GROWING

As its name implies, *region growing* is a procedure that groups pixels or subregions into larger regions based on predefined criteria for growth. The basic approach is to start with a set of "seed" points, and from these grow regions by appending to each seed those neighboring pixels that have predefined properties similar to the seed (such as ranges of intensity or color).

Selecting a set of one or more starting points can often be based on the nature of the problem, as we show later in Example 10.20. When a priori information is not



a b c

FIGURE 10.45 (a) Text image corrupted by sinusoidal shading. (b) Result of global thresholding using Otsu's method. (c) Result of local thresholding using moving averages..

available, the procedure is to compute at every pixel the same set of properties that ultimately will be used to assign pixels to regions during the growing process. If the result of these computations shows clusters of values, the pixels whose properties place them near the centroid of these clusters can be used as seeds.

The selection of similarity criteria depends not only on the problem under consideration, but also on the type of image data available. For example, the analysis of land-use satellite imagery depends heavily on the use of color. This problem would be significantly more difficult, or even impossible, to solve without the inherent information available in color images. When the images are monochrome, region analysis must be carried out with a set of descriptors based on intensity levels and spatial properties (such as moments or texture). We will discuss descriptors useful for region characterization in Chapter 11.

Descriptors alone can yield misleading results if connectivity properties are not used in the region-growing process. For example, visualize a random arrangement of pixels that have three distinct intensity values. Grouping pixels with the same intensity value to form a "region," without paying attention to connectivity, would yield a segmentation result that is meaningless in the context of this discussion.

Another problem in region growing is the formulation of a stopping rule. Region growth should stop when no more pixels satisfy the criteria for inclusion in that region. Criteria such as intensity values, texture, and color are local in nature and do not take into account the "history" of region growth. Additional criteria that can increase the power of a region-growing algorithm utilize the concept of size, likeness between a candidate pixel and the pixels grown so far (such as a comparison of the intensity of a candidate and the average intensity of the grown region), and the shape of the region being grown. The use of these types of descriptors is based on the assumption that a model of expected results is at least partially available.

Let: $f(x, y)$ denote an input image; $S(x, y)$ denote a *seed* array containing 1's at the locations of seed points and 0's elsewhere; and Q denote a *predicate* to be applied at each location (x, y). Arrays f and S are assumed to be of the same size. A basic region-growing algorithm based on 8-connectivity may be stated as follows.

See Sections 2.5 and 9.5 regarding connected components, and Section 9.2 regarding erosion.

1. Find all connected components in $S(x, y)$ and reduce each connected component to one pixel; label all such pixels found as 1. All other pixels in S are labeled 0.
2. Form an image f_Q such that, at each point (x, y), $f_Q(x, y) = 1$ if the input image satisfies a given predicate, Q, at those coordinates, and $f_Q(x, y) = 0$ otherwise.
3. Let g be an image formed by appending to each seed point in S all the 1-valued points in f_Q that are 8-connected to that seed point.
4. Label each connected component in g with a different region label (e.g.,integers or letters). This is the segmented image obtained by region growing.

The following example illustrates the mechanics of this algorithm.

EXAMPLE 10.20: Segmentation by region growing.

Figure 10.46(a) shows an 8-bit X-ray image of a weld (the horizontal dark region) containing several cracks and porosities (the bright regions running horizontally through the center of the image). We illustrate the use of region growing by segmenting the defective weld regions. These regions could be used in applications such as weld inspection, for inclusion in a database of historical studies, or for controlling an automated welding system.

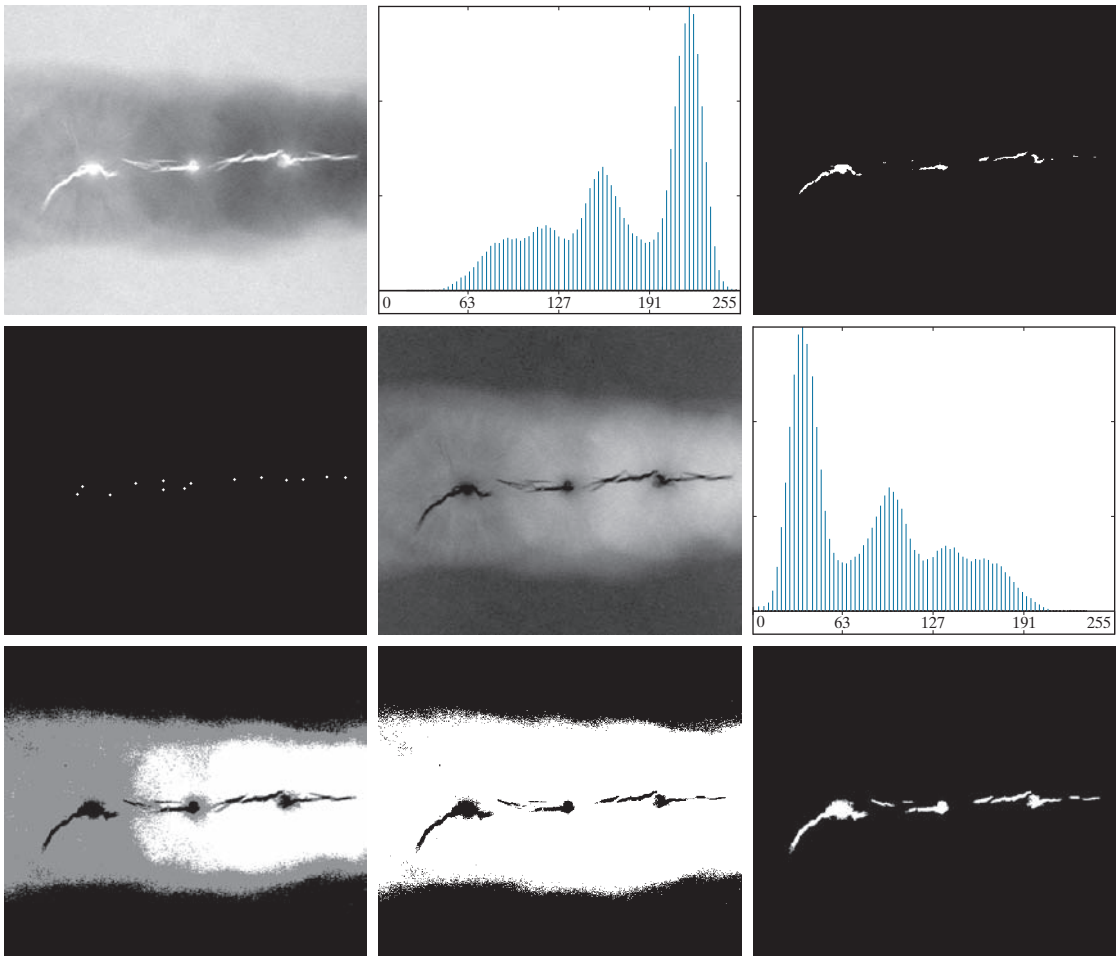
The first thing we do is determine the seed points. From the physics of the problem, we know that cracks and porosities will attenuate X-rays considerably less than solid welds, so we expect the regions containing these types of defects to be significantly brighter than other parts of the X-ray image. We can extract the seed points by thresholding the original image, using a threshold set at a high percentile. Figure 10.46(b) shows the histogram of the image, and Fig. 10.46(c) shows the thresholded result obtained with a threshold equal to the 99.9 percentile of intensity values in the image, which in this case was 254 (see Section 10.3 regarding percentiles). Figure 10.46(d) shows the result of morphologically eroding each connected component in Fig. 10.46(c) to a single point.

Next, we have to specify a predicate. In this example, we are interested in appending to each seed all the pixels that (a) are 8-connected to that seed, and (b) are "similar" to it. Using absolute intensity differences as a measure of similarity, our predicate applied at each location (x, y) is

$$Q = \begin{cases} \text{TRUE} & \text{if the absolute difference of intensities between the seed and the pixel at } (x, y) \text{ is } \leq T \\ \text{FALSE} & \text{otherwise} \end{cases}$$

where T is a specified threshold. Although this predicate is based on intensity differences and uses a single threshold, we could specify more complex schemes in which a different threshold is applied to each pixel, and properties other than differences are used. In this case, the preceding predicate is sufficient to solve the problem, as the rest of this example shows.

From the previous paragraph, we know that all seed values are 255 because the image was thresholded with a threshold of 254. Figure 10.46(e) shows the difference between the seed value (255) and Fig. 10.46(a). The image in Fig. 10.46(e) contains all the differences needed to compute the predicate at each location (x, y). Figure 10.46(f) shows the corresponding histogram. We need a threshold to use in the predicate to establish similarity. The histogram has three principal modes, so we can start by applying to the difference image the dual thresholding technique discussed in Section 10.3. The resulting two thresholds in this case were $T_1 = 68$ and $T_2 = 126$, which we see correspond closely to the valleys of the histogram. (As a brief digression, we segmented the image using these two thresholds. The result in



a b c
d e f
g h i

Figure 10.46 (a) X-ray image of a defective weld. (b) Histogram. (c) Initial seed image. (d) Final seed image (the points were enlarged for clarity). (e) Absolute value of the difference between the seed value (255) and (a). (f) Histogram of (e). (g) Difference image thresholded using dual thresholds. (h) Difference image thresholded with the smallest of the dual thresholds. (i) Segmentation result obtained by region growing. (Original image courtesy of X-TEK Systems, Ltd.)

Fig. 10.46(g) shows that segmenting the defects cannot be accomplished using dual thresholds, despite the fact that the thresholds are in the deep valleys of the histogram.)

Figure 10.46(h) shows the result of thresholding the difference image with only T_1. The black points are the pixels for which the predicate was TRUE; the others failed the predicate. The important result here is that the points in the good regions of the weld failed the predicate, so they will not be included in the final result. The points in the outer region will be considered by the region-growing algorithm as

candidates. However, Step 3 will reject the outer points because they are not 8-connected to the seeds. In fact, as Fig. 10.46(i) shows, this step resulted in the correct segmentation, indicating that the use of connectivity was a fundamental requirement in this case. Finally, note that in Step 4 we used the same value for all the regions found by the algorithm. In this case, it was visually preferable to do so because all those regions have the same physical meaning in this application—they all represent porosities.

REGION SPLITTING AND MERGING

The procedure just discussed grows regions from seed points. An alternative is to subdivide an image initially into a set of disjoint regions and then merge and/or split the regions in an attempt to satisfy the conditions of segmentation stated in Section 10.1. The basics of region splitting and merging are discussed next.

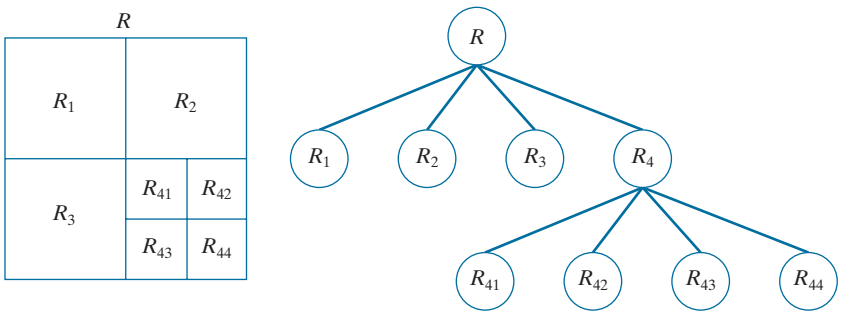
Let R represent the entire image region and select a predicate Q. One approach for segmenting R is to subdivide it successively into smaller and smaller quadrant regions so that, for any region R_i, $Q(R_i) = \text{TRUE}$. We start with the entire region, R. If $Q(R) = \text{FALSE}$, we divide the image into quadrants. If Q is FALSE for any quadrant, we subdivide that quadrant into sub-quadrants, and so on. This splitting technique has a convenient representation in the form of so-called *quadtrees*; that is, trees in which each node has exactly four descendants, as Fig. 10.47 shows (the images corresponding to the nodes of a quadtree sometimes are called *quadregions* or *quadimages*). Note that the root of the tree corresponds to the entire image, and that each node corresponds to the subdivision of a node into four descendant nodes. In this case, only R_4 was subdivided further.

If only splitting is used, the final partition normally contains adjacent regions with identical properties. This drawback can be remedied by allowing *merging* as well as splitting. Satisfying the constraints of segmentation outlined in Section 10.1 requires merging only adjacent regions whose combined pixels satisfy the predicate Q. That is, two adjacent regions R_j and R_k are merged only if $Q(R_j \cup R_k) = \text{TRUE}$.

See Section 2.5 regarding region adjacency.

The preceding discussion can be summarized by the following procedure in which, at any step, we

1. Split into four disjoint quadrants any region R_i for which $Q(R_i) = \text{FALSE}$.
2. When no further splitting is possible, merge any adjacent regions R_j and R_k for which $Q(R_j \cup R_k) = \text{TRUE}$.



a b

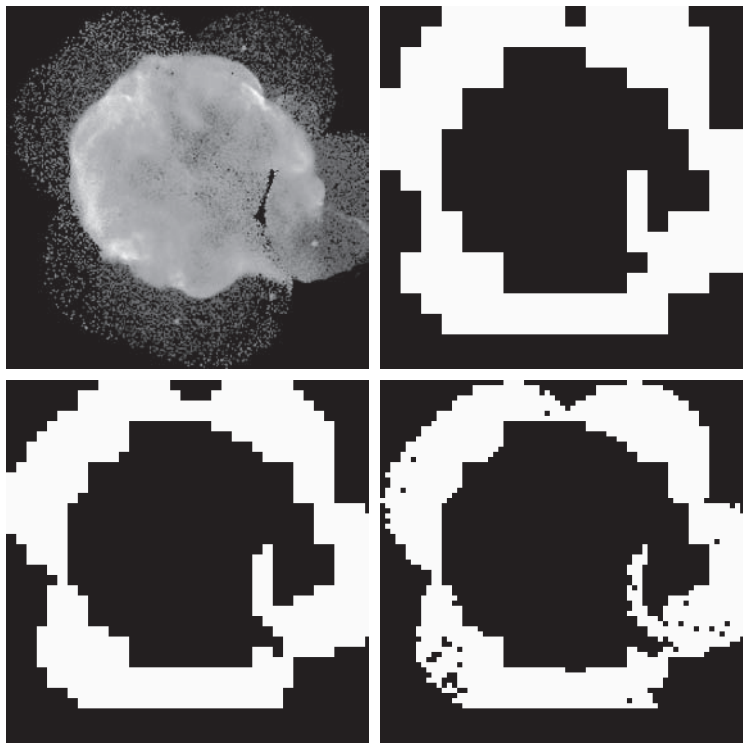
FIGURE 10.47 (a) Partitioned image. (b) Corresponding quadtree. R represents the entire image region.

3. Stop when no further merging is possible.

Numerous variations of this basic theme are possible. For example, a significant simplification results if in Step 2 we allow merging of any two adjacent regions R_j and R_k if each one satisfies the predicate individually. This results in a much simpler (and faster) algorithm, because testing of the predicate is limited to individual quadregions. As the following example shows, this simplification is still capable of yielding good segmentation results.

EXAMPLE 10.21: Segmentation by region splitting and merging.

Figure 10.48(a) shows a 566×566 X-ray image of the Cygnus Loop supernova. The objective of this example is to segment (extract from the image) the "ring" of less dense matter surrounding the dense inner region. The region of interest has some obvious characteristics that should help in its segmentation. First, we note that the data in this region has a random nature, indicating that its standard deviation should be greater than the standard deviation of the background (which is near 0) and of the large central region, which is smooth. Similarly, the mean value (average intensity) of a region containing data from the outer ring should be greater than the mean of the darker background and less than the mean of the lighter central region. Thus, we should be able to segment the region of interest using the following predicate:



a b
c d

FIGURE 10.48
(a) Image of the Cygnus Loop supernova, taken in the X-ray band by NASA's Hubble Telescope. (b) through (d) Results of limiting the smallest allowed quadregion to be of sizes of 32×32, 16×16, and 8×8 pixels, respectively. (Original image courtesy of NASA.)

$$Q(R) = \begin{cases} \text{TRUE} & \text{if } \sigma_R > a \text{ AND } 0 < m_R < b \\ \text{FALSE} & \text{otherwise} \end{cases}$$

where σ_R and m_R are the standard deviation and mean of the region being processed, and a and b are nonnegative constants.

Analysis of several regions in the outer area of interest revealed that the mean intensity of pixels in those regions did not exceed 125, and the standard deviation was always greater than 10. Figures 10.48(b) through (d) show the results obtained using these values for a and b, and varying the minimum size allowed for the quadregions from 32 to 8. The pixels in a quadregion that satisfied the predicate were set to white; all others in that region were set to black. The best result in terms of capturing the shape of the outer region was obtained using quadregions of size 16×16. The small black squares in Fig. 10.48(d) are quadregions of size 8×8 whose pixels did not satisfy the predicate. Using smaller quadregions would result in increasing numbers of such black regions. Using regions larger than the one illustrated here would result in a more "block-like" segmentation. Note that in all cases the segmented region (white pixels) was a connected region that completely separates the inner, smoother region from the background. Thus, the segmentation effectively partitioned the image into three distinct areas that correspond to the three principal features in the image: background, a dense region, and a sparse region. Using any of the white regions in Fig. 10.48 as a mask would make it a relatively simple task to extract these regions from the original image (see Problem 10.43). As in Example 10.20, these results could not have been obtained using edge- or threshold-based segmentation.

As used in the preceding example, properties based on the mean and standard deviation of pixel intensities in a region attempt to quantify the texture of the region (see Section 11.3 for a discussion on texture). The concept of texture segmentation is based on using measures of texture in the predicates. In other words, we can perform texture segmentation by any of the methods discussed in this section simply by specifying predicates based on texture content.

10.5 REGION SEGMENTATION USING CLUSTERING AND SUPERPIXELS

In this section, we discuss two related approaches to region segmentation. The first is a classical approach based on seeking clusters in data, related to such variables as intensity and color. The second approach is significantly more modern, and is based on using clustering to extract "superpixels" from an image.

REGION SEGMENTATION USING K-MEANS CLUSTERING

A more general form of clustering is *unsupervised clustering*, in which a clustering algorithm attempts to find a meaningful set of clusters in a given set of samples. We do not address this topic, as our focus in this brief introduction is only to illustrate how *supervised clustering* is used for image segmentation.

The basic idea behind the clustering approach used in this chapter is to partition a set, Q, of observations into a specified number, k, of clusters. In k-means clustering, each observation is assigned to the cluster with the nearest mean (hence the name of the method), and each mean is called the *prototype* of its cluster. A *k-means algorithm* is an iterative procedure that successively refines the means until convergence is achieved.

Let $\{\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_Q\}$ be set of vector observations (samples). These vectors have the form

$$\mathbf{z} = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix} \tag{10-84}$$

In image segmentation, each component of a vector $\mathbf{z}$ represents a numerical pixel attribute. For example, if segmentation is based on just grayscale intensity, then $\mathbf{z} = z$ is a scalar representing the intensity of a pixel. If we are segmenting RGB color images, $\mathbf{z}$ typically is a 3-D vector, each component of which is the intensity of a pixel in one of the three primary color images, as we discussed in Chapter 6. The objective of k-means clustering is to partition the set Q of observations into k ($k \leq Q$) disjoint cluster sets $C = \{C_1, C_2, \ldots, C_k\}$, so that the following criterion of optimality is satisfied:[†]

$$\arg\min_{C} \left(\sum_{i=1}^{k} \sum_{\mathbf{z} \in C_i} \|\mathbf{z} - \mathbf{m}_i\|^2 \right) \tag{10-85}$$

where $\mathbf{m}_i$ is the *mean vector* (or *centroid*) of the samples in set C_i and $\|\arg\|$ is the vector norm of the argument. Typically, the Euclidean norm is used, so the term $\|\mathbf{z} - \mathbf{m}_i\|$ is the familiar *Euclidean distance* from a sample in C_i to mean $\mathbf{m}_i$. In words, this equation says that we are interested in finding the sets $C = \{C_1, C_2, \ldots, C_k\}$ such that the *sum of the distances* from each point in a set to the mean of that set is minimum.

Unfortunately, finding this minimum is an NP-hard problem for which no practical solution is known. As a result, a number of heuristic methods that attempt to find approximations to the minimum have been proposed over the years. In this section, we discuss what is generally considered to be the "standard" k-means algorithm, which is based on the Euclidean distance (see Section 2.6). Given a set $\{\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_Q\}$ of vector observation and a specified value of k, the algorithm is as follows:

These initial means are the initial cluster centers. They are also called *seeds*.

1. ***Initialize the algorithm:*** Specify an initial set of means, $\mathbf{m}_i(1)$, $i = 1, 2, \ldots, k$.
2. ***Assign samples to clusters:*** Assign each sample to the cluster set whose mean is the closest (ties are resolved arbitrarily, but samples are assigned to only *one* cluster):

   $$\mathbf{z}_q \rightarrow C_i \quad \text{if} \quad \|\mathbf{z}_q - \mathbf{m}_i\|^2 < \|\mathbf{z}_q - \mathbf{m}_j\|^2 \quad j = 1, 2, \ldots, k \; (j \neq i); \quad q = 1, 2, \ldots, Q$$

3. ***Update the cluster centers (means):***

   $$\mathbf{m}_i = \frac{1}{|C_i|} \sum_{\mathbf{z} \in C_i} \mathbf{z} \qquad i = 1, 2, \ldots, k$$

   where $|C_i|$ is the number of samples in cluster set C_i.
4. ***Test for completion:*** Compute the Euclidean norms of the differences between the mean vectors in the current and previous steps. Compute the residual error, E, as the sum of the k norms. Stop if $E \leq T$, where T a specified, nonnegative threshold. Else, go back to Step 2.

[†] Remember, $\min_x(h(x))$ is the minimum of h with respected to x, whereas $\arg\min_x(h(x))$ is the value (or values) of x at which h is minimum.

a b

FIGURE 10.49
(a) Image of size 688×688 pixels. (b) Image segmented using the k-means algorithm with $k = 3$.

When $T = 0$, this algorithm is known to converge in a finite number of iterations to a local minimum. It is not guaranteed to yield the global minimum required to minimize Eq. (10-85). The result at convergence does depend on the initial values chosen for $\mathbf{m}_i$. An approach used frequently in data analysis is to specify the initial means as k randomly chosen samples from the given sample set, and to run the algorithm several times, with a new random set of initial samples each time. This is to test the "stability" of the solution. In image segmentation, the important issue is the value selected for k because this determines the number of segmented regions; thus, multiple passes are rarely used.

EXAMPLE 10.22: Using k-means clustering for segmentation.

Figure 10.49(a) shows an image of size 688×688 pixels, and Fig. 10.49(b) is the segmentation obtained using the k-means algorithm with $k = 3$. As you can see, the algorithm was able to extract all the meaningful regions of this image with high accuracy. For example, compare the quality of the characters in both images. It is important to realize that the entire segmentation was done by clustering of a single variable (intensity). Because k-means works with vector observations in general, its power to discriminate between regions increases as the number of components of vector $\mathbf{z}$ in Eq. (10-84) increases.

REGION SEGMENTATION USING SUPERPIXELS

The idea behind *superpixels* is to replace the standard pixel grid by grouping pixels into primitive regions that are more perceptually meaningful than individual pixels. The objectives are to lessen computational load, and to improve the performance of segmentation algorithms by reducing irrelevant detail. A simple example will help explain the basic approach of superpixel representations.

Figure 10.50(a) shows an image of size 600×800 (480,000) pixels containing various levels of detail that could be described verbally as: "This is an image of two large carved figures in the foreground, and at least three, much smaller, carved figures resting on a fence behind the large figures. The figures are on a beach, with